# Progress on Turyn's converse: general symmetry theorems and coverage below 10,000

Jeffery Kline

21 October 2026
Version 0.1.0 — release candidate

**Abstract**

A symmetric Turyn pair of odd length $t$ is a pair of symmetric $\pm 1$ sequences, one with a zero at the origin, whose periodic autocorrelations add to $q = 2t - 1$ at shift zero and to zero elsewhere. Turyn constructed such pairs whenever $q$ is a prime power and asked whether the converse holds; it remains open. We prove restrictions on the symmetries a pair can have and combine them with classical multiplier theorems. The two principal results are these. If a prime $p \geq 7$ divides $t$ exactly once, no pair is fixed by the decimations that act as squares on the $p$-component and trivially on the rest. On any group $H \times C_3$ with $H$ abelian of odd order, no pair is fixed by inverting the $C_3$ factor unless $H$ is trivial. Both apply when $t$ is squarefree, where earlier conductor tests say nothing. Together with exact quotient tests, norm-lattice certificates, earlier results, and a ray-class transport that excludes an infinite family $q = 13p$, they exclude 655 of the 663 non-prime-power $q < 10000$ with $q \equiv 1 \pmod 4$ that are sums of two squares. Eight remain open, the smallest 3965; a mod-16 identity reduces that case from 198 to 132 unknown signs without settling it. For each result we say what is new and what specializes known methods. Priority was checked against a bounded set of sources only.

## Contents

**1 Introduction** **2**
1.1 What this paper contains . . . 3
1.2 Objects and conventions . . . 3
1.3 The coefficient doubling identity . . . 4

**2 Multiplier, averaging, and prime-fibre obstructions** **5**
2.1 The common multiplier input . . . 5
2.2 Odd-index averaging and the prime-power bound . . . 6
2.3 Full-unit fibres and Ramanujan energy . . . 7
2.4 The index-two prime-fibre theorem . . . 8
2.5 Odd-index prime fibres . . . 10
2.6 Relation to established methods . . . 10

**3 Multiplier kernels and full conductor** **10**

**4 Exact quotient energy and coupled doubling cycles** **12**
4.1 An exact run-count reduction . . . 14

**5 A uniform obstruction to inversion of a three-element factor** **15**

**6 Exact norm lattices and arithmetic transport** **18**
6.1 The period trace form and small-norm rigidity . . . 19
6.2 A complete exact enumeration domain . . . 20
6.3 Factor catalogs and prescribed residue transformations . . . 20
6.4 Ray classes transport complete catalogs . . . 21

**7 A quadratic mod-16 criterion and forced reconstruction** **22**
7.1 The divided correlation defect . . . 22
7.2 The finite-field reconstruction theorem . . . 23
7.3 The exact 3965 consequence and the next gap . . . 25

**8 Prior work and novelty assessment** **25**
8.1 Earlier work on the same problem . . . 26
8.2 Result-by-result comparison . . . 26
8.3 Mann-type tests . . . 27
8.4 Global inversion and forbidden multipliers . . . 27
8.5 Sources we could not check . . . 28

**9 Coverage, certificates, and reproducibility** **28**
9.1 Current unresolved parameters . . . 29
9.2 What the exact checks establish . . . 30

**A Retained auxiliary results and limits** **30**

**B Evidence map** **31**

# 1 Introduction

This paper follows [3], which excluded many composite orders below 2000 and set up the quotient framework used here. That paper is not amended. Everything in Sections 2–7 is new relative to it; Section 8 compares each result with the older literature. Section 1.1 below summarizes what is proved, what is a finite certificate, and what is new.

The finite statement is the following. It rests on the union of the criteria and certificates listed in Section 9, not on one theorem.

**Theorem 1.1** (Recorded coverage below 10000)**.** *Let*

$$\mathcal{E} = \{q : 5 \le q < 10000,\ q \equiv 1 \pmod 4,\ q \text{ is not a prime power and is a sum of two squares}\}.$$

*Then* $|\mathcal{E}| = 663$*. The unconditional arguments and exact certificates listed in Section 9 exclude symmetric cyclic Turyn pairs at 655 members of* $\mathcal{E}$*. They do not decide the remaining eight:*

$$3965, \quad 4745, \quad 5185, \quad 5525, \quad 6409, \quad 7565, \quad 7685, \quad 8245.$$

These eight are open relative to the arguments collected here, not necessarily in the whole literature. The exclusion of 5185 known so far assumes GRH and is not counted. Theorem 1.1 concerns this specific two-circulant construction, not arbitrary Hadamard matrices.

Within all 2499 parameters $q = 5, 9, \ldots, 9997$, there are 642 prime powers, where Turyn's construction gives existence; 1194 fail the necessary two-squares condition; 655 have the further exclusions above; and eight remain unresolved. These categories are disjoint. The degenerate parameter $q = 1$ is outside the stated range.

## 1.1 What this paper contains

Most results take one form: a symmetry is shown to be impossible for any pair, and a classical multiplier theorem then forces that symmetry at particular $q$. T4 and the $C_3$ theorem need only a prime that divides $t$ exactly once, so they apply to squarefree $t$, where the full-conductor test of Section 3 holds automatically and says nothing. Table 1 lists the results, their kind, and how each relates to earlier work. "New" means we did not find the statement in the sources we checked (Section 8); it is not a claim of first publication.

Table 1: Results, their kind, and their relation to earlier work.

| Result | Kind | Relation to earlier work |
|---|---|---|
| Common multiplier group $H_0$ (D1) | theorem | Specialization of Delsarte–Goethals–Seidel [4]; not new. |
| Kernel injectivity; full conductor $F(4t, q) = 4t$; cofactor threshold | theorems | Refines the scalar field-descent bound; new as stated. |
| Odd-index bound O; all-index prime-power bound T2 | theorems | Classical averaging; the nonvanishing step of T2 is new. |
| Full-unit fibre bound T1 | theorem | Orbit-energy method of Tan [14]; new as stated. |
| **Index-two prime-fibre theorem T4** | theorem | **Principal result**; new. |
| Odd-index fibre bound T5 | theorem | Short corollary of standard projection counting. |
| **Inverting a $C_3$ factor on $H \times C_3$** | theorem | **Principal result**; new. |
| Quotient energy QE; doubling-cycle tests | exact criteria | Classical contraction method [9, 15, 16]; the criteria as used here are specific to this problem. |
| Norm-lattice certificates; colored catalogs; ray-class transport | certificates; theorem | Classical lattice and class-field tools; the transport of a certified obstruction is new. |
| Mod-16 identity; forced-fibre reconstruction | theorems | New as stated; prior-art search incomplete. |
| Coverage below 10000 (Theorem 1.1) | finite record | 644 exclusions by the uniform criteria above, 11 by earlier or case-specific certificates. These count coverage, not historically new exclusions. |

We do not claim: the converse; any infinite family of excluded $q$ other than the ray-class family $q = 13p$ and the classical three-power family $t = 3^{4n+1}$ (Corollary 3.3); that the eight listed orders are open in the whole literature; or an unconditional result at 5185, where only a GRH-conditional exclusion is known. Proofs and finite checks were written with AI assistance and checked by separate AI reviewers and separately written programs. That is not expert peer review.

## 1.2 Objects and conventions

Use additive notation for an odd finite abelian group $G$, let 0 be its identity, let $\delta$ denote the group-ring identity, and put $J_G = \sum_{g \in G} g$. For an integer $m$ coprime to the exponent of $G$, $\sigma_m$ sends $g$ to $mg$. The involution $X \mapsto X^{(-1)}$ inverts group elements. On Gaussian group rings the adjoint $X^*$ also conjugates coefficients. A real row is *symmetric* when $X^{(-1)} = X$.

**Definition 1.2.** A symmetric Turyn pair on $G$, $|G| = t$ odd, consists of $A, B \in \mathbb{Z}[G]$ with $A_0 = 0$,

$A_g \in \{-1, 1\}$ for $g \neq 0$, every $B_g \in \{-1, 1\}$, $A^{(-1)} = A$, $B^{(-1)} = B$, and

$$A^2 + B^2 = (2t - 1)\delta = q\delta. \tag{1}$$

The original converse concerns the cyclic group $G = C_t$.

Turyn proved existence when $q$ is a prime power and suggested the converse [2]. Throughout, "order $q$" refers to this parameter; the row length is $t = (q + 1)/2$, the corresponding negacyclic conference matrix has order $q + 1$, and the associated Hadamard construction has order $2(q + 1)$. A nonexistence conclusion for the pair does not transfer to all Hadamard matrices of that order.

Putting $Z = A + iB$, symmetry turns (1) into $ZZ^* = q\delta$. Every character therefore has $|\chi(Z)|^2 = q$. Augmentation gives

$$X^2 + Y^2 = q, \qquad X = A(1) \text{ even}, \quad Y = B(1) \text{ odd}. \tag{2}$$

If $\pi : G \to Q$ has odd fibre size $f$, its projected rows satisfy the same norm equation. The projected $A$ origin is even and has absolute value at most $f-1$; all other projected entries of $A$, and all entries of $B$, are odd and bounded by $f$. Symmetry improves the origin condition to $\bar{A}_0 \equiv f - 1 \pmod 4$. Quotient solutions need not lift to full sign rows.

### 1.3 The coefficient doubling identity

**Lemma 1.3.** *Every pair on an odd finite abelian group of order $t$ satisfies*

$$B_g = (-1)^{(t-1)/2} B_0 A_g A_{2g} \qquad (g \neq 0). \tag{3}$$

*Conversely, for any symmetric $A$ with the prescribed alphabet, choosing $B_0 \in \{-1, 1\}$ and defining $B$ by this rule gives $A^2 + B^2 \equiv q\delta \pmod 8$ coefficientwise.*

*Proof.* Write $A = J_G - \delta - 2a$, $B = J_G - 2b$, where $a, b$ are binary supports and $a_0 = 0$, and let $c = (t - 1)/2$. Expansion of (1) gives

$$a^2 + a + b^2 = c\delta + \ell J_G, \qquad \ell = a(1) + b(1) - c.$$

In characteristic two, squaring doubles group elements. The coefficient at $2g$ is $a_g + a_{2g} + b_g = c[g = 0] + \ell$. At $g = 0$ it gives $b_0 = c + \ell$. Subtracting the latter relation at a nonzero $g$ and returning to signs proves (3). For the converse negate the whole $B$ row if necessary to take $B_0 = 1$. Both binary supports then have even size. Put $K = J_G - \delta$ and $D = \sigma_2$. The sign rule reads $b = (c \bmod 2)K + a + D^{-1}a$ in $\mathbb{F}_2[G]$. The exact defect is

$$4\big[a^2 + b^2 + a + (c - |a| - |b|)J_G - c\delta\big].$$

Its bracket vanishes modulo two by Frobenius, the support parity and the displayed expression for $b$. This proves the converse. □

This exact consequence is used by several later results. Its appearance in a new application is not itself a claim of historical novelty.

## 2 Multiplier, averaging, and prime-fibre obstructions

Throughout this section, $t \geq 3$ is odd, $q = 2t-1$, and $A, B \in \mathbb{Z}[C_t]$ are a symmetric Turyn pair. The identity group element is $\delta$. For real coefficients, $*$ inverts group elements; for Gaussian coefficients it also conjugates coefficients. Thus

$$A_0 = 0, \qquad A_j \in \{-1, 1\}\ (j \neq 0), \qquad B_j \in \{-1, 1\}, \qquad A^* = A, \quad B^* = B, \qquad A^2 + B^2 = q\delta.$$

Write $\sigma_u(\sum_j c_j g^j) = \sum_j c_j g^{uj}$ for decimation by a unit $u$. Every multiplier used below fixes both rows coefficientwise. Signed multipliers, which may negate one row, need not satisfy this hypothesis.

### 2.1 The common multiplier input

For every divisor $m > 1$ of $t$, let $A^{[m]}, B^{[m]}$ be the images under the quotient map $C_t \longrightarrow C_m$. Define

$$\mathcal{H}(m) = \bigcap_{r \mid q \text{ prime}} \langle r^{e_r}, -1 \rangle \ \leq (\mathbb{Z}/m\mathbb{Z})^\times, \qquad e_r = \begin{cases} 1, & r \equiv 1 \pmod 4, \\ 2, & r \equiv 3 \pmod 4. \end{cases} \tag{4}$$

Put $\mathcal{H}_0 = \mathcal{H}(t)$. The image of $\mathcal{H}_0$ modulo $m$ is contained in $\mathcal{H}(m)$, possibly strictly, so the quotient tests may use the larger locally computed group. The groups in the intersection need not be cyclic when $m$ has several prime factors.

**Proposition 2.1** (Common coefficient multipliers; D1)**.** *Every $u \in \mathcal{H}(m)$ fixes $A^{[m]}$ and $B^{[m]}$ coefficientwise.*

*Proof.* This specializes classical common-multiplier reasoning; see the Delsarte–Goethals–Seidel multiplier theorems [4, Theorems 5.2–5.4]. Set $X = A^{[m]} + iB^{[m]}$. Projection preserves symmetry and the correlation identity, so $XX^* = q\delta$, and inversion without coefficient conjugation fixes $X$.

By the Chinese remainder theorem $u$ has a unique lift $\widetilde{u}$ modulo $4m$ with $\widetilde{u} \equiv 1 \pmod 4$. Let $s$ fix $i$ and invert the $m$th roots of unity. Restricting $\langle r, s \rangle$ to automorphisms fixing $i$ replaces $r$ by $r^{e_r}$, so $u \in \mathcal{H}(m)$ if and only if $\widetilde{u} \in \bigcap_{r \mid q} \langle r, s \rangle$.

At a character of order $d \mid m$, let $\alpha$ be the value of $X$ in $\mathbb{Q}(i, \zeta_d)$. Then $\alpha\overline{\alpha} = q$ and $s$ fixes $\alpha$. The primes dividing $(\alpha)$ lie above primes $r \mid q$, which are unramified since $\gcd(q, 4m) = 1$. In this abelian cyclotomic extension the Frobenius automorphism $\sigma_r$ fixes each prime ideal above $r$, so it preserves the $r$-primary part of $(\alpha)$; $s$ preserves it because $s(\alpha) = \alpha$. Hence $(\sigma_{\widetilde{u}}\alpha) = (\alpha)$, and $\sigma_{\widetilde{u}}(\alpha)/\alpha$ is an algebraic integer.

Consider

$$E = q^{-1}\sigma_u(X)X^* \in \mathbb{Q}(i)[C_m].$$

Every character value of $E$, at every character order including the trivial character, is the preceding algebraic integer. By Fourier inversion $mE_j$ is an algebraic integer in $\mathbb{Q}(i)$, so $mE_j \in \mathbb{Z}[i]$. By definition $qE_j \in \mathbb{Z}[i]$. As $\gcd(m, q) = 1$, Bezout's identity gives $E_j \in \mathbb{Z}[i]$. Moreover $EE^* = \delta$, whose identity coefficient is $\sum_j |E_j|^2 = 1$. Hence $E = \varepsilon g^a$ with $\varepsilon$ a Gaussian unit. Inversion without conjugation fixes $E$, so $2a = 0$ in $C_m$, and $a = 0$ since $m$ is odd. The augmentation of $E$ is 1, so $\varepsilon = 1$. Finally $\sigma_u(X) = EX = X$. □

The proof uses every character order. Invariance at primitive characters alone would not give coefficientwise invariance on a composite quotient.

## 2.2 Odd-index averaging and the prime-power bound

Set $X^{[m]} = A^{[m]} + iB^{[m]}$ and $f = t/m$. At each nonzero position the coefficient of $X^{[m]}$ has odd real and imaginary parts, each of absolute value at most $f$. At the origin the real part is even with absolute value at most $f-1$, and the imaginary part is odd with absolute value at most $f$. Every character of $C_m$ takes a value of modulus $\sqrt{q}$ on $X^{[m]}$. Let $\omega(m)$ be the number of distinct prime divisors of $m$.

**Theorem 2.2** (Odd-index averaging; O)**.** *Suppose $H \le (\mathbb{Z}/m\mathbb{Z})^\times$ fixes $X^{[m]}$ and has odd index $d$. Then*

$$m^2 \le 2^{2\omega(m)-1} d^2 q. \tag{5}$$

*In particular one may take $H = \mathcal{H}(m)$ when its index in $(\mathbb{Z}/m\mathbb{Z})^*$ is odd.*

*Proof.* Average the unit action:

$$Y = \frac{1}{\varphi(m)} \sum_{u \in (\mathbb{Z}/m\mathbb{Z})^\times} \sigma_u(X^{[m]}).$$

$H$ has $d$ orbits on the unit positions. If $S$ is the sum of one coefficient from each orbit, then $Y_1 = S/d$. Since $d$ is odd and each summand has two odd coordinates, $|S| \ge \sqrt{2}$.

The value of $Y$ at a character of exact order $e \mid m$ depends only on $e$; call it $Z_e$. It is an average of values of modulus $\sqrt{q}$, so $|Z_e| \le \sqrt{q}$. Fourier inversion at position 1 and the Ramanujan identity $c_e(1) = \mu(e)$ give

$$\frac{mS}{d} = \sum_{e \mid m} \mu(e) Z_e.$$

At most $2^{\omega(m)}$ terms are nonzero. Thus $m\sqrt{2}/d \le 2^{\omega(m)}\sqrt{q}$. No norm identity for $Y$ is needed. □

At even index the parity argument alone allows $S = 0$. For a prime-power quotient this can be ruled out.

**Theorem 2.3** (Prime-power quotient bound at every index; T2)**.** *Let $m = \ell^a \mid t$, where $\ell$ is an odd prime and $a \ge 1$. Let $H$ fix $X^{[m]}$, and put $d = \varphi(m)/|H|$, $f_\ell = t/\ell$, and*

$$c_d = \begin{cases} \sqrt{2}, & d \text{ odd}, \\ 2, & d \text{ even}, \end{cases} \qquad \beta_\ell = \sqrt{(f_\ell - 1)^2 + f_\ell^2}.$$

*Then*

$$c_d |H| \le \min\left\{ 2\sqrt{q}\left(1 - \frac{1}{\ell}\right),\ \sqrt{q} + \beta_\ell \right\}. \tag{6}$$

*Equivalently, the first bound is $c_d m/d \le 2\sqrt{q}$. Hence $m^2 \le d^2 q$ for even $d$, and $m^2 \le 2d^2 q$ for odd $d$.*

*Proof.* Let $S$ again be the sum over the $d$ unit-orbit coefficients, and write $Z_1 = X(1)$. Summing the unit coefficients gives

$$|H| S = Z_1 - X_0^{[\ell]}. \tag{7}$$

The omitted coefficients are exactly those at original indices divisible by $\ell$, so the origin term is that of the $C_\ell$ quotient, not the $C_{\ell^a}$ quotient.

The averaging identity of the preceding proof now has only two nonzero Ramanujan terms:

$$\frac{mS}{d} = Z_1 - Z_\ell, \qquad Z_\ell = \frac{1}{\ell - 1} \sum_{\substack{\chi \text{ on } C_m \\ \operatorname{ord} \chi = \ell}} \chi(X^{[m]}). \tag{8}$$

If $S = 0$, the average on the right equals $Z_1$, which has modulus $\sqrt{q}$. Every term in the average has the same modulus, so equality in the triangle inequality forces each term to equal $Z_1$. These terms are exactly the nontrivial character values of $X^{[\ell]}$, and its trivial character value is also $Z_1$. Fourier inversion would give $X^{[\ell]} = Z_1\delta$. This is impossible, because each coefficient of $X^{[\ell]}$ at a nonzero position has two odd coordinates. Therefore $S \neq 0$.

The coordinates of $S$ have the parity of $d$, so $|S| \geq c_d$: at even index $S$ is a nonzero element of $2\mathbb{Z}[i]$. The triangle inequality applied to (8) gives the first bound. Applied to (7), with $|X_0^{[\ell]}| \leq \beta_\ell$, it gives the second. □

*Remark* 2.4 (Sharpness and scope). At $t = 5$, $q = 9$, $m = 5$, let $\chi_5$ be the quadratic-character element and $J_5$ the all-ones element of $\mathbb{Z}[C_5]$. The identities $\chi_5^2 = 5\delta - J_5$ and $(J_5 - 2\delta)^2 = J_5 + 4\delta$ show that $A = \chi_5$, $B = J_5 - 2\delta$ is a Turyn pair. The subgroup of squares in $(\mathbb{Z}/5\mathbb{Z})^*$ fixes both rows. It has order 2 and index 2, and the second ("box") bound $\sqrt{q} + \beta_\ell$ is attained: $4 = 3 + 1$. The nonvanishing lemma needs a prime-power quotient: at $q = 29$ there is a pair with $S = 0$ on the composite quotient $m = 15$. The parameter $f_\ell = t/\ell$ in the box bound is essential.

### 2.3 Full-unit fibres and Ramanujan energy

Suppose $t = hm$ with $\gcd(h, m) = 1$ and $m > 1$. In the decomposition $C_t = C_h \times C_m$, the subgroup

$$\mathcal{F}_m = \{u \in (\mathbb{Z}/t\mathbb{Z})^\times : u \equiv 1 \pmod{h}\}$$

acts trivially on $C_h$ and by all units on $C_m$.

**Theorem 2.5** (Full-unit fibre bound; T1)**.** *If both rows are fixed by $\mathcal{F}_m$, and $w = \omega(m)$, then*

$$2hm - 1 \leq (2h - 1)4^w + (2^w - 1)^2. \tag{9}$$

*Thus a violation of this inequality, together with $\mathcal{F}_m \leq \mathcal{H}_0$, excludes a pair.*

*Proof.* For each divisor $d \mid m$, let $e_d = \sum_{\gcd(k,m)=d} g_m^k$. The unit group is transitive on each such index set, and $e_m = \delta$. Therefore

$$A = \sum_{d|m} \alpha_d \otimes e_d, \qquad B = \sum_{d|m} \gamma_d \otimes e_d,$$

where the coefficients of $\alpha_d, \gamma_d \in \mathbb{Z}[C_h]$ are signs except for $(\alpha_m)_0 = 0$. At any primitive character $\psi$ of $C_m$, $\psi(e_d) = \mu(m/d)$. Define

$$E = \sum_{d|m} \mu(m/d)\alpha_d, \qquad F = \sum_{d|m} \mu(m/d)\gamma_d.$$

For every character $\phi$ of $C_h$, the pair equation at $\phi \otimes \psi$ gives $|\phi(E)|^2 + |\phi(F)|^2 = q$. Parseval on $C_h$ then gives $\|E\|^2 + \|F\|^2 = q$.

Exactly $2^w$ terms survive in the defining sums, so every coefficient of $E, F$ has absolute value at most $2^w$, except that $|E_0| \leq 2^w - 1$ because $A_0 = 0$. Summing the squared bounds proves (9). □

For an odd coprime split, the inequality fails for every $m > 1$ except $m = 3$, and $m = 15$ with $h \geq 7$ coprime to 15. Indeed, for $w = 1$ it fails exactly when $m \geq 5$; for $w = 2$ it fails when $m \geq 17$, and when $m = 15, h = 1$. For $w \geq 3$, the product of the first $w$ odd primes exceeds $4^w$, which suffices. These are the cases the bound leaves open, not existence claims. The split $h = 3, m = 15$ is inadmissible because it is not coprime.

**Corollary 2.6** (Quotient version of T1)**.** *Let $t' \mid t$, $W = t/t'$, and $t' = h'm$ with $\gcd(h', m) = 1$. If the projected rows on $C_{t'}$ are invariant under $\{u \bmod t' : u \equiv 1 \pmod{h'}\}$, then*

$$q \leq (2h' - 1)4^{\omega(m)}W^2 + (2^{\omega(m)}W - 1)^2.$$

*Proof.* The projected coefficients have absolute value at most $W$, except that the real origin coefficient has absolute value at most $W - 1$. The proof of Theorem 2.5 applies with these bounds; in particular the reduced real origin coefficient is at most $2^{\omega(m)}W - 1$. □

For example, $q = 4453 = 61 \cdot 73$ has $t = 2227 = 131 \cdot 17$. Since $73 \equiv -61^{-1} \pmod{2227}$, $\mathcal{H}_0 = \langle 61, -1 \rangle$. The orders of 61 modulo 17 and 131 are 16 and 5. The elements $61^{5j}$ fix the $C_{131}$ coordinate and, as $\gcd(5, 16) = 1$, exhaust the units on $C_{17}$. Thus the full 17-fibre is forced, whereas (9) would require $4453 \leq 1045$.

### 2.4 The index-two prime-fibre theorem

Let $t = hp$ with $p$ an odd prime and $\gcd(h, p) = 1$. Write $\mathcal{F}_p$ for the prime fibre, and $\mathcal{F}_p^{(2)}$ for its square subgroup: it fixes $C_h$ and multiplies the $C_p$ coordinate by a quadratic residue. On $C_p$, put

$$J = \sum_{k \in C_p} g_p^k, \qquad \chi = \sum_{k \neq 0} \left(\frac{k}{p}\right) g_p^k.$$

**Theorem 2.7** (Index-two prime-fibre obstruction; T4)**.** *If $p \geq 7$, no symmetric Turyn pair of length $hp$ is fixed by $\mathcal{F}_p^{(2)}$. Consequently, a pair is excluded whenever*

$$[\mathcal{F}_p : \mathcal{H}_0 \cap \mathcal{F}_p] \leq 2 \qquad (p \geq 7,\ v_p(t) = 1).$$

*There is no upper bound on $h$ in this statement.*

*Proof.* Invariance makes each $C_p$ fibre constant on $\{0\}$, the nonzero squares, and the nonsquares. Write these three values for $A$ as $\alpha_j, \beta_j, \gamma_j$, and set

$$S_j = \frac{\beta_j + \gamma_j}{2}, \qquad D_j = \frac{\beta_j - \gamma_j}{2}, \qquad U_j = \alpha_j - S_j.$$

Exactly one of $S_j, D_j$ is a sign and the other is zero. Define $S', D', V$ for $B$ in the same way, with $V_j = \alpha'_j - S'_j$. Then

$$A = U \otimes \delta + D \otimes \chi + S \otimes J, \qquad B = V \otimes \delta + D' \otimes \chi + S' \otimes J. \tag{10}$$

Here $\alpha_0 = 0$, every other $\alpha_j$ is a sign, and every $\alpha'_j$ is a sign.

Let $\epsilon = (-1/p)$. Summing the quadratic character gives

$$J^2 = pJ, \qquad \chi J = 0, \qquad \chi^* = \epsilon\chi, \qquad \chi\chi^* = p\delta - J.$$

Symmetry of the full rows gives $U^* = U$, $S^* = S$, $D^* = \epsilon D$, and likewise for $B$. Since $\delta, \chi, J$ are linearly independent, expanding the correlation identity and comparing their coefficients gives, in $\mathbb{Z}[C_h]$,

$$UU^* + VV^* + p(DD^* + D'D'^*) = q\delta, \tag{11}$$

$$UD + VD' = 0. \tag{12}$$

The raw $\chi$-coefficient is $2(UD + VD')$; dividing by 2 is valid in this torsion-free group ring.

Let $D_A = \operatorname{supp} D$ and $D_B = \operatorname{supp} D'$, with sizes $n_A, n_B$. Call the complementary positions type $S$, and let $M = 2h - n_A - n_B$ be the total number of type-$S$ positions in the two rows. We bound $M$ from above and below.

For the upper bound, let $k_A$ count nonzero type-$S$ positions of $A$ where $\alpha_j = -S_j$, and let $k_B$ count type-$S$ positions of $B$ where $\alpha'_j = -S'_j$. Put $k = k_A + k_B$, and let $\eta = 1$ if 0 is type $S$ for $A$, $\eta = 0$ otherwise. The identity coefficient of (11) is

$$\|U\|^2 + \|V\|^2 + p(n_A + n_B) = 2hp - 1.$$

At nonzero type-$D$ positions $U_j^2 = 1$; at nonzero type-$S$ positions it is 0 or 4. At the origin it is 1 if $\eta = 1$ and 0 otherwise. Thus

$$\|U\|^2 + \|V\|^2 = n_A + n_B - 1 + 2\eta + 4k, \qquad (p+1)M = 2h + 2\eta + 4k.$$

Because $k \le M - \eta$,

$$(p-3)M \le 2h - 2\eta \le 2h. \tag{13}$$

For the lower bound, reduce (12) in $\mathbb{F}_2[C_h]$. The sign alphabet gives $\overline{U} = \overline{D} + \delta$ and $\overline{V} = \overline{D'}$, so

$$(\overline{D})^2 + \overline{D} + (\overline{D'})^2 = 0.$$

Squaring in this commutative ring of characteristic two doubles positions. Since $h$ is odd, doubling is a bijection, and the equation becomes the support identity

$$D_A = 2(D_A \triangle D_B). \tag{14}$$

Taking cardinalities gives $|D_A \cap D_B| = n_B/2$. Inclusion–exclusion then gives $n_A + n_B/2 \le h$, so $n_A + n_B \le 3h/2$. As $h$ is odd,

$$M \ge (h+1)/2. \tag{15}$$

Combining (13) and (15) gives $(p-7)h + (p-3) \le 0$, impossible for $p \ge 7$.

For the arithmetic consequence, $\mathcal{F}_p$ is cyclic, so a subgroup of index at most two contains its square subgroup. Proposition 2.1 then supplies the forbidden coefficient symmetry. □

The threshold is necessary: the pair at $p = t = 5$, $q = 9$ in Remark 2.4 is fixed by the square subgroup. In that example the nonsquares negate $A$ and fix $B$, which shows why the sign-$+1$ hypothesis is needed. Hence the exact row-fixing stabilizer of any pair meets a prime fibre with $p \ge 7$ in index at least three. This constrains primes occurring only once in $t$, where the full-conductor condition gives no restriction.

The energy estimate alone is not enough. Without the support identity, (13) gives only a bound that depends on the cofactor $h$. The mod-2 equation forces a positive proportion of type-$S$ positions, which makes the prime threshold independent of $h$.

### 2.5 Odd-index prime fibres

For $k \mid p-1$, let $K_k \le \mathcal{F}_p$ be the unique subgroup of index $k$, and put $f = (p-1)/k$. Its nonzero orbits in a $C_p$ fibre are the $k$ multiplicative cosets, each of size $f$.

**Theorem 2.8** (Odd-index prime-fibre bound; T5)**.** *Let $t = hp$ with $\gcd(h,p) = 1$. If $k$ is odd and a symmetric Turyn pair of length $t$ is fixed by $K_k$, then*

$$f \le k+1, \qquad p \le k^2+k+1. \tag{16}$$

*Consequently a pair is excluded if $k = [\mathcal{F}_p : \mathcal{H}_0 \cap \mathcal{F}_p]$ is odd and $p > k^2+k+1$.*

*Proof.* Project the rows to $C_h$ by summing each $C_p$ fibre. The projected coefficients are

$$\overline{A}_j = \alpha_j + f s_j, \qquad \overline{B}_j = \alpha'_j + f s'_j,$$

where $s_j, s'_j$ are sums of $k$ signs. They are odd, so have absolute value at least one. Thus $|\overline{A}_j| \ge f-1$ for $j \ne 0$, $|\overline{A}_0| \ge f$, and $|\overline{B}_j| \ge f-1$ everywhere. The identity coefficient of the projected correlation equation gives

$$2h(kf+1) - 1 = \sum_j (\overline{A}_j^2 + \overline{B}_j^2) \ge (2h-1)(f-1)^2 + f^2.$$

Expanding and dividing by $2f > 0$ gives $h(f-k-2) \le -1$. Integrality yields $f \le k+1$, which proves both claims. The arithmetic consequence follows from Proposition 2.1. □

This gives $p \le 3$ at index one, $p \le 13$ at index three, and $p \le 31$ at index five. The boundary $p = 13$, $k = 3$ is attained at $h = 1$, $q = 25$: exact enumeration gives twelve symmetric pairs, all fixed by $K_3 = \{1, 5, 8, 12\}$. At $f = k+1$ the slack in the displayed energy bound is $2(h-1)(k+1)$, so attainment at $h = 1$ does not settle $h > 1$.

For example, at $q = 4141 = 41 \cdot 101$ and $t = 2071 = 19 \cdot 109$, $\mathcal{H}_0 \cap \mathcal{F}_{19}$ has order six and index three. Since $19 > 13$, Theorem 2.8 applies. This fibre index must not be replaced by the index of the image of $\mathcal{H}_0$ modulo 19: the fibre subgroup also fixes the other factor, while the image need not.

### 2.6 Relation to established methods

D1 is classical multiplier input. The orbit averaging in O and T2, the Ramanujan reduction in T1, and the projection and second-moment argument in T5 are established methods. The prime-power nonvanishing step and the even-index constant $c_d = 2$ extend the older odd-index estimate; they are not a new multiplier theorem. In T4 the decisive combination is the Legendre decomposition, the support identity (14), and the opposing bounds on $M$. Related decomposition, energy, and characteristic-two arguments occur separately in the literature.

## 3 Multiplier kernels and full conductor

This section constrains the actual coefficient stabilizer, not an enlarged local group, by character cancellation and Parseval followed by the classical common-power multiplier theorem.

**Theorem 3.1** (No principal-unit kernel)**.** *Let $t = hp^a$ be odd, with $p$ prime and $(h,p) = 1$. For $1 \le b < a$, a symmetric Turyn pair on $C_t$ cannot be fixed by every unit satisfying $u \equiv 1 \pmod{h}$ and $u \equiv 1 \pmod{p^b}$. Consequently the exact common coefficient stabilizer $H$ injects under reduction modulo $\mathrm{rad}(t)$, and $|H| \mid \varphi(\mathrm{rad}(t))$.*

*Proof.* Set $L = p^{a-b}$ and $N = hp^b$, so $t = NL$. The kernel consists of the units $u = 1 + Nz$, $z \bmod L$. If $v_p(j) = s < a - b$, the orbit of $j \bmod t$ is $j + hp^{b+s}\mathbb{Z} \bmod t$, of length $p^{a-b-s}$; indices divisible by $L$ are fixed. A character of full $p$-primary order $p^a$ annihilates every nonfixed orbit, so its value on a kernel-invariant row is its value on the subsequence $C'_k = C_{Lk}$, $k \bmod N$.

The two subsequences have total coefficient energy $2N - 1$. Each of the $h\varphi(p^b)$ characters of $C_N$ of full $p$-primary order $p^b$ lifts to such a character, so its two squared magnitudes sum to $q$. Parseval over *all* characters of $C_N$ gives

$$h\varphi(p^b)q \le N(2N-1), \qquad (p-1)q \le p(2N-1).$$

But $q = 2NL-1 \ge 2Np-1$, and $(p-1)(2Np-1)-p(2N-1) = 2Np(p-2)+1 > 0$, a contradiction.

For the corollary, the full reduction kernel is a product of cyclic groups of orders $p^{a-1}$ over the distinct primes dividing $t$. A nontrivial intersection with $H$ contains an element of prime order $p$. Its other prime-primary components are trivial, and its $p$ component generates the unique order-$p$ principal-unit subgroup. This is the forbidden kernel with $b = a-1$, so the intersection is trivial. □

Define, for $p^a \| t$,

$$B_p = \max_{r \mid q} \left\{ v_p(r^{p-1} - 1) + v_p(\mathrm{ord}_{4\,\mathrm{rad}(t)}(r)) \right\}, \qquad F(4t, q) = 4 \prod_{p^a \| t} p^{\min(a, B_p)}.$$

This is the field-descent function in the present coprime odd setting, with the conventions of Tan's Definition 2.10 [14].

**Theorem 3.2** (Full-conductor necessity)**.** *A cyclic symmetric Turyn pair necessarily satisfies* $F(4t, 2t-1) = 4t$.

*Proof.* Fix $p^a \| t$, let $h = t/p^a$, and put $d_r = \mathrm{ord}_{4h}(r)$. Passing from a prime $\ell \ne p$ to $\ell^j$ multiplies a multiplicative order only by an $\ell$-power, and the modulo-$p$ order divides $p - 1$. Hence $v_p(d_r) = v_p(\mathrm{ord}_{4\,\mathrm{rad}(t)}(r))$. If $B_p < a$, odd-prime lifting shows that $r^{(p-1)d_r}$ generates, modulo $p^a$, all principal units congruent to one modulo $p^{b_r}$, where $b_r = v_p(r^{p-1}-1)+v_p(d_r)$. It is one modulo $4h$. Therefore every residue congruent to one modulo $4h$ and $p^{B_p}$ is a power of every prime divisor $r$ of $q$.

To apply the common-power theorem of Delsarte–Goethals–Seidel [4, Theorems 5.2–5.4], work in $\mathbb{Z}[x]/(x^{2t} + 1)$ and put $y = -x^2$, $z = x^t$. Then $y^t = 1$, $z^2 = -1$, and

$$c(x) = A(y) + zB(y), \qquad c(x)c(x^{-1}) = q.$$

The two terms have disjoint even and odd exponent supports, so $c$ is a normalized negacyclic conference polynomial with one zero coefficient. The cited theorem makes each common power $u$ a multiplier: $c(x^u) = \varepsilon x^k c(x)$. The unique zero coefficient forces $k = 0$. Since $u \equiv 1 \pmod 4$, $z^u = z$, so $A(y^u) = \varepsilon A$ and $B(y^u) = \varepsilon B$; as $B(1)$ is odd, $\varepsilon = 1$. Substitution fixes $z$ and sends $y$ to $y^u$, so separating the parity supports fixes both rows coefficientwise. Reduction modulo $t$ supplies the kernel forbidden by Theorem 3.1, with $b = B_p \ge 1$. Thus $B_p \ge a$ for every $p$, proving the assertion. The 2-part of the field-descent function is four: in Tan's definition the modulus is divisible by 4 and every prime $r \mid q$ is odd. □

This statement concerns the arithmetic function $F$, not the claim that every individual character value has minimal conductor $4t$. For squarefree $t$ it is automatic.

**Corollary 3.3** (Explicit three-primary threshold)**.** *Write $t = h3^a$, $a \geq 1$, $(h, 3) = 1$, and $e(h) = \max(\{0\} \cup \{v_3(p-1) : p \mid h\})$. If a pair exists and $q = 2t - 1$ is composite, then*

$$2 \cdot 3^a \leq h3^{2e(h)}. \tag{17}$$

*In particular, at $t = 3^a$ and at $t = 5 \cdot 3^a$, existence forces $q$ prime.*

*Proof.* Put $\beta = \max_{r|q} v_3(r^2 - 1)$. The 3-part of the unit-group exponent modulo $4h$ is at most $3^{e(h)}$, so full conductor gives $a \leq \beta + e(h)$. Choose a prime $r \mid q$ attaining $\beta$ and let $s = q/r > 1$. For $M = 3^{\min(a,\beta)}$, since $q \equiv -1 \pmod{M}$, the factors $r, s$ lie in opposite classes $\pm 1 \pmod{M}$. Both are odd and greater than one, even if not coprime, so their minimum sizes are $2M - 1$ and $2M + 1$. Thus $2h3^a = q + 1 \geq 4M^2 \geq 4 \cdot 3^{2a-2e(h)}$, which is (17). For $h = 1, 5$, $e(h) = 0$ and the inequality fails for every $a \geq 1$. □

For example, $h = 7, 13$ have $e(h) = 1$ and force primality for $a \geq 4$, while $h = 11$ does so for $a \geq 2$. The class $t = 3^{4n+1}$, $n \geq 1$, is excluded unconditionally: $q = 2 \cdot 3^{4n+1} - 1$ is divisible by five and exceeds five.

Older scalar field-descent bounds also give eventual primality for fixed $h$. The two formulations differ: at $q = 305$, $t = 153$, $F(612, 305) = 204$ and $\varphi(204) = 64$, so the scalar inequality $305 \leq 204^2/64$ passes although full conductor fails.

## 4 Exact quotient energy and coupled doubling cycles

Projection to a quotient turns a sign pair into integer rows: the identity coefficient gives an energy equation, and the other coefficients give the full correlation equations. Exact multiplier orbits make both reductions finite and exact for their stated relaxations. The doubling identity (3) then couples the two rows.

Let $M \leq (\mathbb{Z}/t\mathbb{Z})^\times$ contain $-1$ and fix both rows. Its orbits on $\mathbb{Z}/t\mathbb{Z}$ are $O_0 = \{0\}, O_1, \ldots, O_{N-1}$. Write $a_i, b_i$ for the row values on $O_i$, so $a_0 = 0$ and every other variable is a sign. The applications below use the guaranteed common multiplier group $\mathcal{H}_0$. For $m \mid t$, let $I_m$ be the image of $M$ modulo $m$, and let $\mathcal{C}_m$ be its orbits on $\mathbb{Z}/m\mathbb{Z}$. We call these quotient classes. Write $\pi_m$ for the induced group-ring projection.

**Theorem 4.1** (Exact quotient-energy criterion)**.** *Every orbit $O_i$ projects onto one quotient class $c(i)$, and its fibre size*

$$N_i = \#\{x \in O_i : x \equiv j \pmod{m}\} = \frac{|O_i|}{|c(i)|} \qquad (j \in c(i))$$

*is independent of $j$. Define the finite signed-sum sets*

$$\mathcal{S}_c^A = \left\{ \sum_{\substack{i : c(i) = c \\ i \neq 0}} N_i a_i : a_i \in \{-1, 1\} \right\}, \qquad \mathcal{S}_c^B = \left\{ \sum_{i : c(i) = c} N_i b_i : b_i \in \{-1, 1\} \right\}.$$

*If a symmetric Turyn pair exists, there are $X_c \in \mathcal{S}_c^A$, $Y_c \in \mathcal{S}_c^B$ satisfying*

$$\sum_{c \in \mathcal{C}_m} |c|(X_c^2 + Y_c^2) = q. \tag{18}$$

*Conversely, every solution of (18) in these finite sets is realized by symmetric $M$-invariant alphabet rows whose projected energy is $q$. Thus absence of such a solution excludes a pair.*

*Proof.* The projection is equivariant for $M \to I_m$, so an orbit maps onto one class. If $j, j'$ lie in that class, some $u \in M$ takes $j$ to $j'$ modulo $m$, and multiplication by $u$ bijects the corresponding fibres inside $O_i$. Hence $\pi_m(A)$ and $\pi_m(B)$ have class values

$$X_c = \sum_{i:c(i)=c} N_i a_i, \qquad Y_c = \sum_{i:c(i)=c} N_i b_i.$$

Projection preserves the group-ring equation, and its identity coefficient is the sum of the squares of all projected coefficients; this is (18). For the converse, each orbit lies in exactly one class, so the signed sums in different classes, and in the two rows, involve disjoint variables and can be chosen independently. Since $-1 \in M$, the resulting rows are symmetric. This converse asserts the energy equation only. □

The complete projected identity is a stronger finite condition. Put

$$M^{(r)}_{cc'} = \#\{j \in c : j + r \in c'\}, \qquad r \in \mathbb{Z}/m\mathbb{Z}.$$

Then every pair satisfies

$$\sum_{c,c' \in \mathcal{C}_m} M^{(r)}_{cc'}(X_c X_{c'} + Y_c Y_{c'}) = q\,[r = 0] \quad \text{for every } r. \tag{19}$$

This groups the correlation sum by the classes of $j$ and $j + r$. We call (18) the quotient-energy condition and (19) the full quotient condition. Their feasible points are marginals; feasibility does not give a Turyn pair.

**Corollary 4.2** (Odd-multiplicity energy floor)**.** *For each of $\mathcal{S}^A_c$ and $\mathcal{S}^B_c$, consider the multiset of fibre weights $N_i$ of the orbits it sums. Let $w$ be the largest weight, let $n$ be its multiplicity, and let $R$ be the sum of all remaining weights. Set*

$$d = \begin{cases} \max(0, w - R), & n \text{ odd}, \\ 0, & n \text{ even}, \end{cases}$$

*and set $d = 0$ for an empty multiset. Denote the two resulting bounds by $d_A(c), d_B(c)$. A pair is impossible if*

$$\sum_c |c|\big(d_A(c)^2 + d_B(c)^2\big) > q.$$

*Proof.* An odd number of signs has an odd, hence nonzero, sum. So the terms of weight $w$ have absolute sum at least $w$, and the remaining terms at most $R$. Apply the triangle inequality and Theorem 4.1. In the $A$-origin class, $O_0$ is excluded from the multiset because its coefficient is zero. □

At $m = 1$, the criterion keeps the exact row sums. If the nonzero orbit sizes are $s_i$ and $g = \gcd_i s_i$, then

$$A(1) \equiv t - 1 \pmod{2g}, \qquad B(1) \equiv t - 1 \pm 1 \pmod{2g}, \qquad A(1)^2 + B(1)^2 = q. \tag{20}$$

Here $A(1), B(1)$ denote augmentation. Counting the orbits of each size sharpens these congruences. At $q = 8281$, the nonzero orbit sizes are two 20's, two 50's, and twenty 200's, so $B(1) \equiv \pm 1 \pmod{20}$. The representations $8281 = 0^2 + 91^2 = 84^2 + 35^2$, with the parity of the two row sums, require $B(1) \in \{\pm 91, \pm 35\}$, a contradiction. At $q = 4685$, the orbit inventory gives $A(1) \equiv \pm 2$

(mod 20) and $B(1) \equiv \pm 1, \pm 3 \pmod{20}$. The only representations are $4685 = 14^2+67^2 = 62^2+29^2$; the first fails the $A$ congruence and the second the $B$ congruence.

The energy floor excludes $q = 9809$ at $m = 9$. In every nonzero quotient class the weights are fifteen 36's, one 4, and one 1; the origin has the same weights for $B$ and omits the 1 for $A$. Thus the energy is at least

$$32^2 + 31^2 + 8 \cdot 2 \cdot 31^2 = 17361 > 9809.$$

The exact criterion also excludes 4705 at $m = 13$ and 4525 at $m = 31$; their finite signed-sum data are in the coverage certificates.

### 4.1 An exact run-count reduction

Normalize $B_0 = 1$ and put $\varepsilon = (-1)^{(t-1)/2}$. Doubling permutes the nonzero $M$-orbits, because it is invertible and commutes with $M$. For a cycle $C$ of this permutation, write $L_C$ for its length and $a_0, \dots, a_{L_C-1}$ for its cyclic sequence of $A$ signs. Let $k_C$ count its negative signs and let $2r_C$ count its changes of sign, including the closing edge.

**Theorem 4.3** (Simultaneous quotient signatures)**.** *Choose any collection of divisors $m \mid t$ such that doubling fixes every class in $\mathcal{C}_m$. Equivalently, $2 \in I_m$, with $m = 1$ interpreted trivially. A doubling cycle lying above a class $c$ has a constant fibre weight, denoted $N_{m,C}$. Proposed simultaneous class coefficients $X_{m,c}, Y_{m,c}$ are realized by symmetric $M$-invariant alphabet rows satisfying the full equation modulo 8 (to which Lemma 1.3 applies, since its proof uses only that congruence) if and only if, for every nonzero doubling cycle, integers $k_C, r_C$ can be chosen with*

$$\begin{cases} r_C = 0, \quad k_C \in \{0, L_C\}, & \text{or} \\ 1 \le r_C \le \lfloor L_C/2 \rfloor, \quad r_C \le k_C \le L_C - r_C, \end{cases} \tag{21}$$

*and, simultaneously for every chosen quotient and class,*

$$X_{m,c} = \sum_{C \text{ above } c} N_{m,C}(L_C - 2k_C), \tag{22}$$

$$Y_{m,c} = [c = \{0\}] + \varepsilon \sum_{C \text{ above } c} N_{m,C}(L_C - 4r_C). \tag{23}$$

*The same variables $k_C, r_C$ occur at all quotients.*

*Proof.* The edge signs $a_i a_{i+1}$ have product 1, so the number of transitions is even. The node and edge sums are

$$\sum_i a_i = L_C - 2k_C, \qquad \sum_i a_i a_{i+1} = L_C - 4r_C.$$

A word without transitions is constant. With $2r_C > 0$ transitions, there are $r_C$ nonempty runs of each sign, which gives (21). Conversely, split $k_C$ into $r_C$ positive run lengths and $L_C - k_C$ into another $r_C$, then alternate the runs. This realizes every allowed pair $(k_C, r_C)$, the constant cases included.

If doubling fixes a quotient class, all orbits in a cycle lie above that class and have the same size, hence the same fibre weight. The coefficient identity $B_j = \varepsilon A_j A_{2j}$ then gives (22)–(23). For sufficiency, choose the cyclic words independently on the disjoint cycles, assign their signs to the $M$-orbits, and define $B$ by this identity. The rows have the prescribed projections and, by Lemma 1.3, satisfy the full equation modulo 8. □

Doubling always fixes the origin class, so for every $m \mid t$ its possible $B$ coefficients are exactly

$$1+\varepsilon\left[\frac{t}{m}-1-4\sum_{C \text{ above } 0} w_C r_C\right], \qquad 0 \le r_C \le \lfloor L_C/2 \rfloor, \tag{24}$$

where $w_C$ is the size of an orbit in $C$. Since every allowed transition count is attained by a cyclic word, (24) is exact for the $M$-invariant mod-8 relaxation. A pair must also satisfy $\pi_m(B)_0^2 \le q$, a finite obstruction from the origin coefficient alone. Equations (22)–(23) with (19) keep the stronger coupling between different quotients.

**Corollary 4.4** (A fixed origin fibre)**.** *Let $h \mid t$, put $m = t/h$, and suppose the image of $M$ modulo $h$ contains 2. Then every pair satisfies*

$$\pi_m(B)_0 = B_0[1+\varepsilon(h-1)], \qquad [1+\varepsilon(h-1)]^2 \le q.$$

*No coprimality assumption on $h$ and $m$ is required.*

*Proof.* Choose $u \in M$ with $u \equiv 2 \pmod h$. For $j$ divisible by $m$, $(u-2)j$ is divisible by $t$, so $A_{2j} = A_{uj} = A_j$. Hence each of the $h-1$ nonzero coefficients of $B$ in that fibre is $\varepsilon B_0$, and the origin adds $B_0$. Apply the projected energy equation. □

For $q = 6437$, $t = 3219 = 3 \cdot 29 \cdot 37$ and

$$M = \langle 41, -1\rangle \cap \langle 157, -1\rangle \le (\mathbb{Z}/3219\mathbb{Z})^\times$$

has order 72. Above the origin of $C_3$, its three nonzero doubling cycles have $(L_C, w_C) = (14, 72), (2, 18), (7, 4)$, of total size 1072, and $\varepsilon = -1$. The first edge contribution $72(14-4r) = 144(7-2r)$ is an odd multiple of 144; the rest lies between $-56$ and 64. So the total edge sum is at least 88 or at most $-80$, and

$$|\pi_3(B)_0| \ge 81, \qquad 81^2 = 6561 > 6437,$$

a contradiction. The orbit inventory is an exact arithmetic certificate.

For $q = 7625$, $t = 3813 = 3 \cdot 31 \cdot 41$ and $M = \langle 5, -1\rangle \cap \langle 61, -1\rangle$. The congruences

$$5^3 \equiv 125 \equiv -61^{19} \pmod{3813}, \qquad 125 \equiv 2 \pmod{123}$$

show that the image of $M$ modulo 123 contains 2. As $\varepsilon = 1$, Corollary 4.4 gives

$$|\pi_{31}(B)_0| = 123, \qquad 123^2 = 15129 > 7625,$$

with no enumeration of full quotient marginals.

## 5 A uniform obstruction to inversion of a three-element factor

The energy method for a full prime fibre has an exceptional boundary at $p = 3$. The next theorem resolves it for every odd finite abelian complementary group, using the complete coefficient equations rather than only their identity coefficients. Throughout this section, $\delta$ is the identity element of the group ring, $J_H = \sum_{g \in H} g$, and $\sigma_a$ is the automorphism induced by multiplication by $a$ when this is invertible on $H$.

**Lemma 5.1** (Quadratic symmetry transfer)**.** *Let $H$ be finite abelian, let $E, \Lambda \in \mathbb{Z}[H]$, and suppose*

$$\Lambda(3\Lambda - 2E) = q\delta, \qquad 3 \nmid q.$$

*Every group automorphism fixing $E$ also fixes $\Lambda$.*

*Proof.* For an automorphism $\sigma$ fixing $E$ and a character $\phi$ of $H$, the algebraic integers $\phi(\Lambda)$ and $\phi(\sigma\Lambda)$ are roots of $3z^2 - 2\phi(E)z - q = 0$. If they were distinct, their product would be $-q/3$, which is not an algebraic integer. So they agree for every character, and character separation gives $\sigma\Lambda = \Lambda$. □

**Lemma 5.2** (Complementary-support rigidity)**.** *Let $H$ have odd order $h$, and suppose $W$ and $2W$ partition $H \setminus \{0\}$. Let $D \in \mathbb{Z}[H]$ have support $W$ and coefficients $\pm 1$ on $W$. If $E = \epsilon\delta + 2D$, where $\epsilon \in \{-1, 1\}$, satisfies $\sigma_2(E)^2 = E^2$, then*

$$E + \sigma_2(E) = 2\epsilon J_H, \qquad E(1) = \epsilon h, \qquad D = \epsilon \mathbf{1}_W.$$

*Proof.* The support partition gives $\sigma_2(D) - D = J_H - \delta + 2Q$ for some $Q \in \mathbb{Z}[H]$. For a nontrivial character $\phi$, equal squares give $\phi(\sigma_2 E) = \pm\phi(E)$. The plus sign would give $0 = -1 + 2\phi(Q)$, contradicting algebraic integrality. So every nontrivial character annihilates $E + \sigma_2 E$, whose coefficients are then constant by Fourier inversion; the identity coefficient is $2\epsilon$. This is the first formula, and augmentation gives the second. At $g \in W$ the first formula gives $2D_g = 2\epsilon$, the third. When $h = 1$, all conclusions are immediate. □

**Theorem 5.3** (Full three-fibre obstruction)**.** *Let $H$ be a finite abelian group of odd order $h$, and put $G = H \times C_3$. Suppose $A, B \in \mathbb{Z}[G]$ are symmetric, $A_0 = 0$, all other $A$ coefficients and every $B$ coefficient are signs, and*

$$A^2 + B^2 = (6h - 1)\delta.$$

*If both rows are fixed by the automorphism $(g, y) \mapsto (g, y^{-1})$, then $h = 1$. At $h = 1$ there are precisely the four pairs*

$$A = (0, s, s), \qquad B = (b, -b, -b), \qquad s, b \in \{-1, 1\}.$$

*Proof.* Write

$$A = U + S(y + y^2), \qquad B = V + T(y + y^2).$$

The four slices are symmetric on $H$, since the two nonzero $C_3$ slices are equal and the rows are symmetric on $G$. The trivial and nontrivial characters of $C_3$ give the full identities

$$E^2 + 4F^2 = q\delta, \qquad E = U - S, \qquad 2F = V - T, \tag{25}$$

$$(U + 2S)^2 + (V + 2T)^2 = q\delta, \qquad q = 6h - 1. \tag{26}$$

Here $E = \epsilon\delta + 2D$, with $\epsilon = -S_0$, $D_0 = 0$, and all coefficients of $D, F$ in $\{0, \pm 1\}$. Let $Z$ be the nonzero zero set of $D$, and let $Z'$ be the full zero set of $F$. The identity coefficient of (25), and (25) divided by 4, give

$$|Z| + |Z'| = \frac{h - 1}{2}, \qquad \epsilon D + D^2 + F^2 = c\delta, \quad c = \frac{3h - 1}{2}. \tag{27}$$

By symmetry, the identity coefficients are sums of squares.

In characteristic 2, squaring doubles subscripts, so (27) reduces to

$$\mathbf{1}_Z + \mathbf{1}_{2Z} + \mathbf{1}_{2Z'} = J_H + (c \bmod 2)\delta. \tag{28}$$

Each of the $h-1$ nonzero elements is covered a positive odd number of times on the left, while the total nonzero coverage is at most

$$2|Z| + |Z'| \le 2(|Z| + |Z'|) = h - 1.$$

Equality forces $Z' = \varnothing$ and makes $Z, 2Z$ a disjoint partition of the nonzero elements. The origin coefficient in (28) forces $c$ odd, so $h \equiv 1 \pmod 4$. Hence

$$V = -T = F, \qquad W := \operatorname{supp} D = 2Z, \qquad 2W = Z, \qquad 4W = W. \tag{29}$$

In particular, $F$ is a full sign row.

Next we determine enough signs of $D$ to obtain a multiplier. Write $h = 4k+1$ and let $N = \{s \in W : \epsilon D_s = -1\}$. For a sign row $X$ supported on $Y$,

$$X^2 = \mathbf{1}_{2Y} + 2P(X),$$

where $P(X)$ sums $X_a X_b$ over unordered distinct pairs with a fixed sum; modulo 2 it is the unsigned pair count $p_Y$. For $Y = H$, every sum has $(h-1)/2 = 2k$ unordered distinct pairs, since doubling is bijective. Reducing (27) modulo 4 therefore gives

$$\mathbf{1}_N + p_W = J_H + (k+1)\delta \quad \text{in } \mathbb{F}_2[H]. \tag{30}$$

For $s \in W$, no diagonal pair in $W^2$ has sum $s$, because $s/2 \notin W$. So, with $c_W(s) = \#\{(a,b) \in W^2 : a+b = s\}$,

$$D_s = \begin{cases} \epsilon, & c_W(s) \equiv 2 \pmod 4, \\ -\epsilon, & c_W(s) \equiv 0 \pmod 4. \end{cases}$$

By (29) the bijection $(a,b) \mapsto (4a, 4b)$ preserves $W^2$, so $c_W(4s) = c_W(s)$ and

$$\sigma_4(E) = E. \tag{31}$$

Put $R = \sum_{s \in Z} S_s s$ in $\mathbb{Z}[H]$ and $\Lambda = 2R - \epsilon\delta$. Coordinatewise,

$$U = \frac{\Lambda + E}{2}, \quad S = \frac{\Lambda - E}{2}, \quad U + 2S = \frac{3\Lambda - E}{2}, \quad V + 2T = -F.$$

Subtracting (25) from four times (26) gives

$$9\Lambda^2 - 6\Lambda E = 3q\delta, \qquad \Lambda(3\Lambda - 2E) = q\delta.$$

Since $3 \nmid q$, Lemma 5.1 and (31) imply $\sigma_4(\Lambda) = \Lambda$ and $\sigma_4(S) = S$.

The coefficient doubling identity holds on every odd finite abelian group: its proof uses only the sign encoding, characteristic-2 Frobenius, and invertibility of doubling. On $G$ at the nonzero coordinate $(s, y)$ the sign factor is $(-1)^{(3h-1)/2} = -1$, so

$$F_s = F_0 S_s S_{2s} \qquad (s \in H).$$

Since $S_{4s} = S_s$, this gives $F_{2s} = F_s$. Applying $\sigma_2$ to (25) then gives $\sigma_2(E)^2 = E^2$, and Lemma 5.2 gives $E(1) = \epsilon h$.

Finally let $f = F(1)$ and $v = (U + 2S)(1)$. Here $f$ is odd because $F$ has $h$ sign coefficients. Augmenting the two slice identities gives

$$6h - 1 = h^2 + 4f^2 = v^2 + f^2.$$

Since $f^2 \geq 1$, $(h-1)(h-5) \leq 0$; with $h \equiv 1 \pmod 4$ this leaves $h = 1$ or $h = 5$. At $h = 5$ the first equality forces $f^2 = 1$ and the second gives $v^2 = 28$, impossible. At $h = 1$ the slice equations give exactly the four stated pairs, and direct substitution verifies them. □

**Corollary 5.4** (Coordinate-free form)**.** *Let $G$ be finite abelian of odd order, and let $\tau$ be an involutive automorphism of $G$ with exactly three elements satisfying $\tau(g) = -g$. A symmetric Turyn pair fixed by $\tau$ can exist only when $|G| = 3$.*

*Proof.* Division by 2 is well-defined on $G$, and

$$g = \frac{g + \tau(g)}{2} + \frac{g - \tau(g)}{2}$$

splits $G$ as the direct product of the fixed and anti-fixed subgroups. Their intersection is trivial, being killed by 2. The anti-fixed subgroup has order 3, so $\tau$ is inversion on a $C_3$ factor and the identity on its complement. Apply Theorem 5.3. □

**Corollary 5.5** (Full prime-fibre obstruction)**.** *Let $p$ be an odd prime and let $H$ be a finite abelian group of odd order $h$. A symmetric Turyn pair on $H \times C_p$ fixed by every automorphism acting identically on $H$ and as a unit on $C_p$ exists only at $p = 3, h = 1$.*

*Proof.* The case $p = 3$ is Theorem 5.3. For $p \geq 5$, unit transitivity on $C_p \setminus \{0\}$ gives

$$A = U + S(J_{C_p} - \delta), \qquad B = V + T(J_{C_p} - \delta).$$

In the projection to $H$, the $A$-origin coefficient has magnitude $p - 1$, and every other $A$ coefficient and every $B$ coefficient has magnitude at least $p - 2$. So the energy is at least

$$(p-1)^2 + (2h-1)(p-2)^2,$$

which exceeds $q = 2ph - 1$ by $2(p-1)[h(p-4)+1] > 0$, a contradiction. □

For cyclic $t = ph$ with $p \nmid h$, the Chinese remainder theorem identifies the relevant automorphisms with the pure $p$-fibre in $(\mathbb{Z}/t\mathbb{Z})^\times$. So a guaranteed common multiplier group containing this fibre excludes the parameter unless $t = 3$. For $p = 3$, the pure fibre has order 2; when $t > 3$, its intersection with the common row stabilizer must be trivial.

Theorem 5.3 contains the earlier support-tiling and mod-16 sign constraints without their finite search boundary, and supersedes a bounded full-three-fibre exclusion for $1 < h \leq 1700$. Because $H$ is an arbitrary odd abelian group, no cyclicity or coprimality assumption on $H$ in $H \times C_3$ is needed; cyclic lengths with $9 \mid t$ are not of this form and are not covered. The $p \geq 5$ part of Corollary 5.5 is the elementary energy argument; the additional content is the $p = 3$ boundary. These results assume extra symmetry; they are not the unrestricted Turyn converse.

## 6 Exact norm lattices and arithmetic transport

When the preceding criteria leave a nontrivial character problem, exact integral lattices give a finite certificate without uncertified class-group decisions. Their factorwise invariants also give an infinite, arithmetically specified family.

### 6.1 The period trace form and small-norm rigidity

Let $\ell$ be an odd prime, $H \le \mathbb{F}_\ell^*$ contain $-1$, put $h = |H|$, $d = (\ell-1)/h$, $K = \mathbb{Q}(\zeta_\ell)^H$, and $L = K(i)$. For a generator $g$ of $\mathbb{F}_\ell^*$ write $\eta_j = \sum_{a \in g^j H} \zeta_\ell^a$, $0 \le j < d$.

**Lemma 6.1.** *The $\eta_j$ form an integral basis of $K$, $\mathcal{O}_L = \mathcal{O}_K[i]$, and*

$$T = (\mathrm{Tr}_{K/\mathbb{Q}}(\eta_j \eta_k)) = \ell I_d - h J_d. \tag{32}$$

*For $\alpha = \sum c_j \eta_j \in \mathcal{O}_K \setminus \mathbb{Z}$, $\mathrm{Tr}(\alpha^2) \ge \ell - h$.*

*Proof.* $H$ permutes the integral basis $\zeta_\ell, \dots, \zeta_\ell^{\ell-1}$, so the orbit sums are a basis of the invariant module $\mathcal{O}_K$. In a product of two coset sums, exactly $h$ ordered pairs sum to zero when the cosets agree, and none otherwise; the full cyclotomic trace is $\ell - 1$ at exponent zero and $-1$ otherwise. Dividing by $h$ gives (32). Its determinant $\ell^{d-1}$ is coprime to the discriminant $-4$ of $\mathbb{Q}(i)$, so the coprime-discriminant compositum theorem gives $\mathcal{O}_L = \mathcal{O}_K[i]$.

The trace form is

$$Q(c) = h \sum_{j<k} (c_j - c_k)^2 + \sum_j c_j^2.$$

Since $\sum \eta_j = -1$, $\alpha$ is rational exactly when all $c_j$ are equal. Otherwise, splitting at an integer between their minimum and maximum gives at least $d-1$ unequal pairs, and $\sum c_j^2 \ge 1$, so $Q(c) \ge h(d-1) + 1 = \ell - h$. □

**Proposition 6.2** (Two rigidity thresholds)**.** *Suppose $\alpha, \beta \in \mathcal{O}_K$ and $\alpha^2 + \beta^2 = N$ for a positive integer $N$. If $dN < \ell - h$, both components are rational integers. The same holds at equality if $d$ is odd. If, in addition, $\alpha + i\beta \equiv 1 \pmod{2\mathcal{O}_L}$, then a stronger sufficient condition is*

$$(d \text{ odd or } N < \ell), \qquad dN < 8(\ell - h) + d - 4.$$

*Proof.* Taking traces gives the strict first bound. At equality, a nonrational component would take the entire trace and make the other zero, so its square would be rational; an odd-degree field has no quadratic subfield.

For the second bound, write the components in the period basis: all real coefficients are odd and all imaginary coefficients even. A nonconstant odd vector costs at least $4h(d-1) + d$ in the trace form, and a nonconstant even vector at least $4h(d-1) + 4$, so two nonconstant components cost at least $8(\ell - h) + d - 4$. If exactly one is nonrational, its square is an integer. Odd degree again rules this out. In even degree the only possible quadratic subfield of the real cyclic field is $\mathbb{Q}(\sqrt{\ell})$; a nonrational integral element with rational square is an integral multiple of $\sqrt{\ell}$, of square at least $\ell$. This contradicts $N < \ell$. □

The degree-or-norm clause is essential. Forced Gaussian ideal factors may first be removed from a character value, provided the residual is proved integral with the right parity. At 8865, with conductor 31, $d = 5$, $h = 6$, removing factors of norms 9 and 197 leaves norm 5, at the endpoint $dN = \ell - h = 25$. The character is Gaussian. After division by three, its odd real coordinate has residues $\{\pm 27, \pm 29\}$ modulo 31, whereas an even real augmentation has residues $\{\pm 12, \pm 16\}$; these sets are disjoint.

### 6.2 A complete exact enumeration domain

**Theorem 6.3** (Trace-minimal generators)**.** *Let $L$ be a CM field of degree $2d$, let $q$ be a positive rational integer, and let $I$ be an integral ideal satisfying $I\overline{I} = (q)$. For every nonzero $x \in I$,*

$$\mathrm{Tr}_{L/\mathbb{Q}}(x\overline{x}) \geq 2dq,$$

*with equality exactly when $x\overline{x} = q$.*

*Proof.* The inclusion $(x) \subseteq I$ gives $|N_{L/\mathbb{Q}}(x)| \geq N(I) = q^d$. Apply the arithmetic–geometric mean inequality to the $d$ positive conjugates of $x\overline{x}$. Equality requires $|N(x)| = q^d$ and equal conjugates, each then equal to $q$. The converse is immediate. □

In the period setting, a full integral basis matrix $B_I$ of $I$ gives the normalized lattice form

$$Q_I = \frac{1}{q} B_I^{\mathsf{T}} \,\mathrm{diag}(T, T) B_I, \qquad \mathrm{rank}\, Q_I = 2d, \quad \det Q_I = \ell^{2d-2}.$$

The form is integral: $x\overline{y}/q \in \mathcal{O}_K[i]$ for $x, y \in I$, so the trace of its real part is an integer. The determinant follows from $|\det B_I| = N(I) = q^d$. By Theorem 6.3, complete enumeration through normalized length $d$ finds exactly the desired norm generators. The determinant and cutoff do not depend on $q$, but the number of allocation ideals can be large.

For $p^e \| q$, allocation on a pair $\mathfrak{p}, \overline{\mathfrak{p}}$ chooses exponents $a, e-a$, $0 \leq a \leq e$; a self-conjugate prime requires $2a = e$. Unique ideal factorization makes this exhaustive. An exported lattice must be proved equal to the whole ideal by membership and exact index; enumerating a proper sublattice would not certify nonexistence.

The $C_{61}$ computations use $H = \{1, 13, 14, 47, 48, 60\}$, $h = 6$, $d = 10$. At 8905, all 16384 allocation ideals give 88 signed norm generators, 44 satisfying parity and eight boxed marginals with $(|X|, |Y|) = (72, 61)$. The complete $C_{73}$ classification has twelve marginals, all with $(|X|, |Y|) = (16, 93)$. A pair would have the same augmentation at both quotients, a contradiction. So a nonempty marginal at one quotient does not defeat the method: compatibility between quotients gives the exclusion.

At $2561 = 13 \cdot 197$, the 1048576 allocations give 504 signed norm generators, 252 satisfying parity; these have real conductor residues

$$\{0, \pm 2, \pm 8, \pm 14, \pm 16, \pm 25, \pm 30\} \pmod{61}.$$

Every permitted augmentation has $(|X|, |Y|) = (40, 31)$ or $(44, 25)$, with real residues $\{\pm 21, \pm 17\}$, disjoint from these. So no integral augmentation-compatible lift exists. These classifications use exact integer and rational arithmetic only; no unfinished class-group certification is a premise.

### 6.3 Factor catalogs and prescribed residue transformations

**Theorem 6.4** (Colored factor matching)**.** *Let $I\overline{I} = (a)$, $J\overline{J} = (b)$ in a CM field, and suppose $\gamma\overline{\gamma} = ab$, $(\gamma) = IJ$. Then multiplication by $\gamma/b$ is an integral-lattice isometry from $\overline{J}$ to $I$, with their respective normalized trace forms. If $a, b$ are odd, $\gamma \equiv 1 \pmod 2$, and a multiplicative residue map $\rho$ that commutes with complex conjugation has $\rho(\gamma) = \alpha$ with $b, \alpha$ invertible, their complete vector counts agree under the color map*

$$(x \bmod 2, \rho(x)) \longmapsto (\overline{x} \bmod 2,\ (\alpha/b)\overline{\rho(x)})$$

*from $J$ to $I$.*

*Proof.* As fractional ideals, $(\gamma/b)\overline{J} = IJ\overline{J}/(b) = I$, so the map is bijective. The multiplier has relative norm $a/b$, which cancels the normalization in the two forms. Its residue is one modulo two and $\alpha/b$ under $\rho$. Composing with conjugation from $J$ to $\overline{J}$ gives the stated color transformation. □

So a mismatch at an exactly enumerated length is an exclusion, while a match on a finite prefix proves nothing about existence. For 2561, each factor has 56 Galois orbits of allocation ideals. Enumerating these 112 factor lattices through normalized length 80 rejects every product allocation under all eight allowed augmentation transformations. An independently organized exact enumeration confirms the mismatch and, with the actual residues, keeps all 252 parity-compatible generators.

### 6.4 Ray classes transport complete catalogs

**Theorem 6.5** (Arithmetic transport)**.** *Let $L/\mathbb{Q}$ be Galois and CM, of degree $2d$, with central complex conjugation $c$. Let $\mathfrak{m} = N\mathcal{O}_L$, and let $p, p_0$ be completely split rational primes coprime to $N$. Choose primes $\mathfrak{P}, \mathfrak{P}_0$ above them with the same ray class modulo $\mathfrak{m}$. Then the entire labeled collections of allocation lattices, with normalized trace forms and ambient residue colors modulo $\mathfrak{m}$, are isometric after a label permutation. If $i \in L$ and $N$ is even, the Gaussian augmentation residues of norms $ap$ and $ap_0$ also agree for fixed $a$ coprime to $pp_0$.*

*Proof.* Ray equality gives $\beta$ with $(\beta) = \mathfrak{P}/\mathfrak{P}_0$ and $\beta \equiv 1 \pmod{\mathfrak{m}}$ locally at the modulus primes. For each set $T$ of one member from each pair $\{\sigma, c\sigma\}$ in $\mathrm{Gal}(L/\mathbb{Q})$, set

$$I_p(T) = \prod_{\sigma \in T} \sigma\mathfrak{P}, \qquad \lambda_T = \prod_{\sigma \in T} \sigma\beta.$$

Then $I_p(T) = \lambda_T I_{p_0}(T)$ and

$$\lambda_T \overline{\lambda_T} = \prod_{\sigma} \sigma\beta = N_{L/\mathbb{Q}}(\beta) = p/p_0.$$

The last equality has no real-unit ambiguity: a CM field norm is positive, and the fractional ideal norm is $p/p_0$. Multiplication by $\lambda_T$ is an integral-module bijection preserving the normalized trace form. It preserves each ambient color, since $\lambda_T \equiv 1$ at the modulus primes and the difference of two integral vectors is integral at every other prime. So the catalogs agree at *every* length, not only up to a cutoff.

For the Gaussian assertion, put $\Delta = N_{L/\mathbb{Q}(i)}(\beta)$. If $\pi_0$ generates the prime below $\mathfrak{P}_0$, then $\pi = \Delta\pi_0$ is a Gaussian integer of norm $p$, with $\pi \equiv \pi_0 \pmod{N}$. Unique factorization writes every norm-$ap$ Gaussian integer as $z\pi$ or $z\overline{\pi}$, where $N(z) = a$; replacing $\pi$ by $\pi_0$ gives the residue-preserving bijection, including the even-real, odd-imaginary parity requirement. □

**Corollary 6.6** (An infinite excluded family)**.** *Take*

$$H = \{1, 13, 14, 47, 48, 60\}, \qquad K = \mathbb{Q}(\zeta_{61})^H, \qquad L = K(i),$$

*and put $\mathfrak{m} = 122\mathcal{O}_L$. Fix a prime $\mathfrak{P}_0$ above 197. Let $\mathcal{S}$ be the rational primes $p \neq 13$, $p \nmid 122$, which split completely in $L$ and have some prime $\mathfrak{P}$ with $[\mathfrak{P}] = [\mathfrak{P}_0]$ in $\mathrm{Cl}_{\mathfrak{m}}(L)$. For every $p \in \mathcal{S}$ there is no cyclic symmetric Turyn pair at $q = 13p$, $t = (13p+1)/2$. The set $\mathcal{S}$ has positive natural density among rational primes.*

*Proof.* Taking norms of the ray congruence gives $p \equiv 197 \pmod{122}$, and splitting gives $p \equiv 1 \pmod 4$. So $61 \mid t$, and the extended local groups generated by $13, -1$ and by $p, -1$ modulo 61 both equal $H$. The multiplier theorem puts the hypothetical primitive character in $L$, with norm $13p$, parity one modulo two, and conductor residue equal to an actual even-real, odd-imaginary augmentation. Theorem 6.5 carries all factor catalogs and augmentation colors to those at $13 \cdot 197$, preserving the joining multiplier $\alpha/p$ modulo 61. The certified empty joins through length 80 then give a contradiction by Theorem 6.4.

Let $M$ be the ray class field of $L$ for $\mathfrak{m}$. The modulus is Galois-stable, so $M/\mathbb{Q}$ is Galois. Let $\mathcal{C}$ be the conjugacy class of the Artin element of $\mathfrak{P}_0$ in $\mathrm{Gal}(M/\mathbb{Q})$. Its restriction to $L$ is trivial. Chebotarev [19, Theorems 28.6 and 28.9] identifies the completely split primes with the required ray class precisely with this Frobenius class, up to finitely many omitted primes. Hence their density is

$$\delta_{\mathcal{S}} = \frac{|\mathcal{C}|}{[M : \mathbb{Q}]} > 0.$$

□

The class-field and Chebotarev inputs are classical; no GRH, numerical class-group completeness or simultaneous-prime hypothesis is used. The resulting $q$ are distinct squarefree composites with counting function asymptotic to $\delta_{\mathcal{S}} \operatorname{Li}(X/13)$: positive density among the varying prime factors, not among all integers. The congruence $p \equiv 197 \pmod{244}$ alone is not claimed to characterize the family.

A separately checked 77-digit prime in this ray state comes with an explicit principal ray identity and ECPP/Pratt certificates. It certifies membership only; it does not compute the ray class group or the density. It lies beyond the range of Theorem 1.1 and adds no row below 10000.

# 7 A quadratic mod-16 criterion and forced reconstruction

The cycle counts of the preceding sections forget where the signs sit within each cycle. The next congruence layer keeps those positions and gives a uniform elimination theorem.

## 7.1 The divided correlation defect

Let $G$ be any odd finite abelian group of order $t > 1$, normalize $B_0 = 1$, and let $P$ be the negative support of $A$. Write $D = \sigma_2$, $K_0 = J_G - \delta$, $c = (t-1)/2$, $e = c \bmod 2$ and $\eta = |P|/2 \bmod 2$. For a binary support $X$, let $W(X)$ be the group-ring sum of unordered pairs of distinct elements of $X$, reduced modulo two. All products below are convolutions.

**Theorem 7.1** (Quadratic mod-16 identity)**.** *Define $B$ by (3). Then the exact integer correlation defect is divisible coefficientwise by eight, and in $\mathbb{F}_2[G]$,*

$$\begin{aligned} F(P) &= W(P) + PD(P) + eP + (1-e)D(P) + D^2(P) \\ &\quad + \big(\lfloor c/2 \rfloor + \eta\big)K_0 + \eta\delta = D\left(\frac{A^2 + B^2 - q\delta}{8}\right). \end{aligned} \tag{33}$$

*Consequently the full mod-16 equations are exactly $F(P) = 0$.*

*Proof.* Put $Q = (1-B)/2$ coefficientwise, $R = D^{-1}P$, $\epsilon = 1 - 2e$, $p_0 = |P|$ and $T_0 = \sum_g P_g R_g$. Both $p_0$ and $T_0$ are even by inversion symmetry. The sign rule gives

$$Q = eK_0 + \epsilon(P + R - 2P \circ R), \quad |Q| = e(t-1) + \epsilon(2p_0 - 2T_0),$$

where $\circ$ is coefficientwise intersection. Integer expansion gives

$$A^2 + B^2 - q\delta = 4[P^2 + Q^2 + P + \alpha J_G - c\delta], \quad \alpha = c - p_0 - |Q|.$$

For a binary vector $X$, $X^2 = DX + 2W_{\mathbb{Z}}(X)$, where $W_{\mathbb{Z}}$ is the unreduced unordered-pair sum. At a nonzero $s$, the remaining numerator $D(P+Q)_s + P_s + \alpha$ is even, with half modulo two equal to

$$\lfloor c/2 \rfloor + p_0/2 + (1-e)(P_{s/2} + P_s) + P_{s/2}P_s.$$

This uses $e + \epsilon c = 2\epsilon\lfloor c/2 \rfloor$ and even $T_0$. At the origin the correlation defect is exactly zero.

For binary $X, Y$, expanding the symmetric difference gives

$$W(X \triangle Y) = W(X) + W(Y) + XY + D(X \circ Y) \quad \text{in } \mathbb{F}_2[G].$$

Apply this to $Q = P \triangle R \triangle eK_0$. Symmetric supports have even weight, so $K_0X = X$ in characteristic two. At nonzero positions $W(K_0)_s = c - 1$, and $e(c-1) = 0 \pmod 2$. Therefore

$$W(P) + W(Q) = D^{-1}W(P) + PR + D(P \circ R) + e(R + DP)$$

away from the origin. The coefficientwise intersection term cancels $P_{s/2}P_s$ in the previous half-numerator. Applying $D$ gives (33) at nonzero positions. At zero, $W(P)_0 = p_0/2$, $(PDP)_0 = T_0 = 0 \pmod 2$, and the added $\eta\delta$ cancels the first term. □

Write $P = \sum_i x_i E_i$ over disjoint even-sized symmetry orbits. The coefficient of $x_i x_j$, $i \neq j$, is $E_iE_j + E_iD(E_j) + E_jD(E_i)$, the linear coefficient is

$$W(E_i) + E_iD(E_i) + eE_i + (1-e)D(E_i) + D^2(E_i) + (|E_i|/2 \bmod 2)J_G,$$

and the constant is $(\lfloor c/2 \rfloor \bmod 2)K_0$. So the criterion is quadratic in the orbit bits. Replacing $A$ by $-A$ leaves $B$ and the defect unchanged.

### 7.2 The finite-field reconstruction theorem

Let $p \equiv 5 \pmod 8$ be prime, $H \leq \mathbb{F}_p^*$ have order $h \equiv 2 \pmod 4$, and $n = (p-1)/h$ be even. Assume that 2 generates $\mathbb{F}_p^*/H$. Let $C \leq H$ have odd order $h/2$. Consider symmetric rows on $C_p \times C_3$ invariant under $C$ on the first coordinate. Normalize $A_{(0,1)} = A_{(0,2)} = 1$ and $B_{(0,0)} = 1$ and use the doubling rule. If $y$ generates $C_3$, then

$$P = U + Vy + V^{(-1)}y^2, \qquad U^{(-1)} = U, \quad U_0 = V_0 = 0.$$

The binary vector $U$ is $H$-invariant and $V$ is $C$-invariant. From here on, a star on a field value denotes the involution induced by inversion on $C_p$.

**Theorem 7.2** (Forced inner fibre)**.** *Choose a primitive pth root $\zeta$ in characteristic two. Put $L_2 = \mathbb{F}_{2^{2n}}$, $K_2 = \mathbb{F}_{2^n}$ and*

$$u = U(\zeta), \quad v = V(\zeta), \quad w = W(V)(\zeta), \quad v^* = v^{2^n}.$$

*The $C$ periods form a basis of $L_2$, and the $H$ periods form a self-dual normal basis of $K_2$. If $v \notin K_2$, every solution of the full mod-16 equations has the unique possible inner value*

$$u = vv^* + \frac{vw + v^*w^*}{v^3 + (v^*)^3}. \tag{34}$$

*This reconstructs one binary $H$-invariant $U$; it need not satisfy the remaining equations. Odd $|V|$ is a sufficient condition for $v \notin K_2$.*

*Proof.* Write $g = h/2$ odd and $\theta_i = \sum_{a \in 2^i C} \zeta^a$, $0 \le i < 2n$. Since 2 generates modulo $H$ and $n$ is even, it generates modulo $C$, and $2^n C = -C$. Frobenius cycles the $2n$ periods, each of absolute trace one. Counting zero and nonzero exponents in their trace products gives

$$\mathrm{Tr}_{L_2/\mathbb{F}_2}(\theta_i \theta_j) = 1 + [j = i + n],$$

because the uniform nonzero multiplicity is $g - [j = i + n]$ and the zero contribution is even. The Gram matrix is $J + R$, where $R$ exchanges paired indices; since $(J + R)^2 = I$ in characteristic two, the periods form a basis. The same count for $\beta_i = \theta_i + \theta_{i+n}$ gives $\mathrm{Tr}_{K_2/\mathbb{F}_2}(\beta_i \beta_j) = [i = j]$. So evaluation is a bijection on the stated binary supports, even when the cyclotomic polynomial is reducible over $\mathbb{F}_2$.

As $g$ is odd, $\mathrm{Tr}_{L_2/\mathbb{F}_2}(v) = |V| \pmod 2$, and elements of $K_2$ have zero trace from $L_2$; this proves the odd-weight guard. Here $t = 3p \equiv 7 \pmod 8$, so the class-$y^2$ component of (33), evaluated at $\zeta$, is

$$v^* u^2 + (v^* + v^2)u = w + v(v^*)^2 + v^* + (v^*)^4. \tag{35}$$

Its unordered-pair component is $W(V) + UV^{(-1)}$; the three product components are $UD_xV$, $V^{(-1)}D_xU$ and $VD_xV^{(-1)}$. The linear terms give the last two terms on the right.

Conjugate (35) and eliminate $u^2$. The determinant is $d_0 = v^3 + (v^*)^3$. If it vanished, $v/v^*$ would be a cube root of unity. All cube roots of unity lie in $K_2$ because $n$ is even, so conjugation would both fix and invert the ratio, forcing it to be one, contrary to $v \notin K_2$. Thus $d_0 \neq 0$. The remaining numerator expands to $vw + v^*w^* + vv^*d_0$, giving (34). The expression is fixed by the involution, so it lies in $K_2$, and the period basis reconstructs exactly one possible $U$. □

**Proposition 7.3** (The complete residual test)**.** *In Theorem 7.2, define*

$$\alpha = 1 + v^2/v^*, \quad \beta = w/v^* + vv^* + 1 + (v^*)^3, \quad a_0 = \alpha + \alpha^*, \quad b_0 = \beta + \beta^*.$$

*Then $a_0 \neq 0$, $u = b_0/a_0$, and semilinear compatibility is equivalent to*

$$b_0^2 + a_0(\alpha^*\beta + \alpha\beta^*) = 0 \quad \text{in } K_2. \tag{36}$$

*The remaining primitive class-zero equation is*

$$W(U)(\zeta) + vv^* + d_0 + u^3 + u + u^4 + 1 = 0. \tag{37}$$

*These two tests and the projected mod-16 equation on $C_3$ are together equivalent to the full mod-16 equation in this normalized sector.*

*Proof.* Divide (35) by $v^*$ to obtain $u^2 + \alpha u = \beta$. Subtract its conjugate. Since $a_0 = d_0/(vv^*) \neq 0$, this forces $u = b_0/a_0$. Multiplication of the substituted residual by $a_0^2$ yields precisely (36). Expanding the class-zero component of (33) gives (37); the constant is one because $t \equiv 7 \pmod 8$.

Each $C$-invariant component vanishing at $\zeta$ vanishes at every nontrivial $p$th root, since $C$ and Frobenius together reach all exponents. In characteristic two, a polynomial of degree at most $p-1$ with these zeros is 0 or $J_{C_p}$; augmentation separates them, since $p$ is odd. So the projected equation excludes $J_{C_p}$ in each component. The other nonzero $C_3$ component is the conjugate of the first, and Theorem 7.1 completes the equivalence. □

Every prime $p \equiv 5 \pmod 8$ admits $h = (p-1)/2$, $n = 2$, because 2 is a quadratic nonresidue, so the reconstruction criterion has infinitely many parameter instances. This is not an infinite family of exclusions, and the needed symmetry is not claimed at every associated order.

### 7.3 The exact 3965 consequence and the next gap

At $q = 3965$, $t = 1983 = 3 \cdot 661$, the common multiplier group has order ten. Its 199 nonzero orbits form doubling cycles of lengths 132, 66 and one. After fixing the last sign, the exact $C_3$ equations and congruences force

$$\bar{A} = (0, 11, 11), \qquad \bar{B} = (61, -1, -1), \qquad (k_{132}, r_{132}) = (65, 33), \quad (k_{66}, r_{66}) = (33, 18);$$

the complete integer enumeration of these quotient possibilities is in the certificates. So $|V| = 325$, $|U| = 330$ for every normalized candidate, and Theorem 7.2 eliminates all 66 inner variables, reducing 198 bits to 132.

For the saved outer support, the forced inner support has weight 310 and fails (35). Independently, the integer correlations give a 132-by-66 affine system of coefficient rank 66 and augmented rank 67; a listed XOR of 35 rows is $0 = 1$. So *every* one of its $2^{66}$ inner fillings is excluded. All 66 relative rotations of the saved two words also fail, while keeping the exact $C_3$ data and full mod-8 equations. Other outer words remain unclassified, so this does not close 3965 and does not change Theorem 1.1.

A cheaper trace filter has a proved limitation. Put the 132-type orbit bits $x_j$ in doubling order, pair indices $i, i + n$, and align the inner bits $y_i$ by the projected $H$ periods. Set $a_i = x_i + x_{i+n}$, $z_i = a_i + y_i$. Specialize the $C_3$ generator to one *before* evaluating and taking the absolute trace from $K_2$. Then

$$\sum_i \big(x_i x_{i+n} + a_i y_i + z_i z_{i+1}\big) = 0. \tag{38}$$

Self-duality makes the cross-pair contribution $\sum_i (x_i x_{i+n} + a_i y_i)$ and the $PDP$ contribution $\sum_i z_i z_{i+1}$; the other trace terms cancel. For a fixed inner weight $k_y$ and run count $r_y$, expansion makes this linear in $y$:

$$\sum_i (a_i + a_{i-1} + a_{i+1}) y_i = k_y - r_y + \sum_i x_i x_{i+n} + \sum_i a_i a_{i+1}.$$

If both word weights are odd and $n$ is even, the left sides summed over all $n$ relative rotations equal one. Since $n$ is even, they cannot all be equal, so the filter accepts some phases and rejects others and cannot exclude an entire pair of word shapes. For the saved 3965 words it accepts 37 phases and rejects 29; all fail the full mod-16 equations.

Whether full mod-16 correlation forces even $|V|$ throughout this normalized arithmetic class is open. At $p = 101, h = 10, n = 10$, two separate exhaustive implementations tested all 524288 odd outer supports: 620 pass semilinear compatibility, and none passes (37). This is a finite result only. A nonzero relative trace to $\mathbb{F}_4$ cannot replace the full class-zero condition: an explicit compatible $p = 101$ support has zero projected defect and zero relative trace but a nonzero class-zero residual; its full integer correlations are in the certificates.

The open problem is to control the weighted unordered-pair terms in the two complete residual equations.

## 8 Prior work and novelty assessment

A checked proof is not a claim of priority. We compared each result with the sources listed below; that search was bounded. Throughout, "new" means that we did not find the statement in those sources. Priority is unresolved for every result in this paper.

## 8.1 Earlier work on the same problem

A symmetric Turyn pair of length $t$ gives a negacyclic conference matrix of order $2t = q + 1$ and a cyclic relative difference set with parameters

$$(m, n, k, \lambda) = (2t, 2, 2t - 1, t - 1)$$

in $C_{4t}$. Delsarte, Goethals and Seidel pose the prime-power conjecture for negacyclic conference matrices in Section 6 of [4]. Jungnickel and Pott treat the corresponding $(q+1, 2, q, (q-1)/2)$ family, the conjecture, and multiplier-based nonexistence searches [9, pp. 344–347]. These are predecessors on the same question. Field descent and weighing-matrix bounds [12, 13, 14] are broader methods that we specialize.

Multiplier theory, character inversion, quotient group rings, intersection numbers [1], Gaussian periods, self-conjugacy and orbit enumeration are established methods, and we claim none of them. The quotient-energy criterion, for instance, is the classical contraction method: projected coefficients are sums over fibres, multiplier orbits constrain those sums, and the identity coefficient of the norm equation is an exact sum of squares. Such equalities and orbit searches appear in [15, 16], and [9] searched this family by multiplier orbits. What is specific here is the two-row formulation, its coupling constraints, and the certificates.

An excluded order need not be newly excluded. For example, $q = 425$ is the period-852 problem settled by Arasu and Voss [10]. The coverage counts in this paper measure what the collected arguments decide, not how many orders are excluded for the first time.

## 8.2 Result-by-result comparison

Table 2 gives, for each result, the closest earlier work we found and our status label, using the same labels as Table 1.

Table 2: Closest earlier work and status of each result.

| Result | Closest earlier work found | Status |
|---|---|---|
| Common multiplier group $H_0$ (D1) | Common-power multiplier theorems with their sign convention [4, Theorems 5.2–5.4]. | Not new: specialization. |
| Doubling identity (Lemma 1.3) | Elementary characteristic-2 computation. | Not claimed as new. |
| Odd-index bound O; 3-adic index formula | Averaging, parity and Ramanujan-sum inversion are classical. | New as stated. |
| Three-power and fixed-cofactor eventual primality | Follow from the published scalar $F$-bound; a second route uses [4]. | Not new: specialization. |
| Kernel injectivity; full conductor; sharp cofactor threshold | Classical multipliers and Parseval. The scalar $F$-bound does not give all applications, and the Mann tests of [8, 9] do not imply it by the substitution in Section 8.3. | New as stated; Arasu–Ma comparison open. |
| Full-unit fibre bound T1; all-index prime-power bound T2 | Orbit reductions and multiplier-order bounds in the proofs of [14, Theorems 4.6 and 4.10]. | New as stated, including T2's nonvanishing step. |
| **Index-two prime-fibre theorem T4** | Legendre decompositions with energy bounds, e.g. [14, Theorem 4.3]; none has the quadratic-residue stabilizer with the mod-2 support identity. | **Principal result; new.** |
| Odd-index fibre bound T5 | Short projection and second-moment argument. | Corollary of standard counting. |

| Result | Closest earlier work found | Status |
|---|---|---|
| Auxiliary extensions (Appendix A) | Extend T2 and T4 under stated hypotheses. | No separate comparison made. |
| Quotient energy QE; finite sign searches | [9, 15, 16]; a related Gaussian two-component form in [18, Section 5]. | Method classical; criteria specific. |
| **Inverting a $C_3$ factor on $H \times C_3$** | Global-inversion theorems of [5] use a different action (Section 8.4); none gives the symmetry-transfer chain. | **Principal result; new.** |
| Doubling-cycle and fixed-fibre tests | Elementary consequences of the doubling identity. | Method elementary; criteria specific. |
| Small-norm rigidity; trace-minimal enumeration | Gaussian-period trace forms and ideal-lattice enumeration. | Specific cutoffs; method classical. |
| Colored catalogs; ray-class transport | Ideal lattices, ray classes, CM norm products and Chebotarev [19]. | New as stated. |
| Mod-16 identity; forced-fibre reconstruction | Gaussian periods and characteristic-2 elimination. | New as stated; prior-art search incomplete. |

Two infinite families of excluded $q$ appear. The three-power family $t = 3^{4n+1}$, $n \geq 1$, follows from the published scalar $F$-bound; we include it as a specialization. The ray-class theorem carries the certified $13 \cdot 197$ obstruction to a set of primes $p$ of positive density among primes, giving excluded $q = 13p$; the set of such $q$ has density zero among integers.

### 8.3 Mann-type tests

Quotient the relative difference set by the odd subgroup of order $t/d$, $d \mid t$. For a prime $r \mid q$, Result 1.8 of [8] gives

$$r^f \equiv -1 \pmod{4d} \implies v_r(q) = 2j \text{ and } r^j \leq t/d,$$

and [9, Result 3.10] allows the trigger $\tau r^f \equiv -1 \pmod{4d}$ for an available multiplier $\tau$. Its other branch is empty here because $q^2 - 4t(t-1) = 1$. These tests overlap the full-conductor criterion but do not imply it by this substitution. At $q = 305 = 5 \cdot 61$, $t = 153$, the multipliers guaranteed by [9, Result 3.8] are generated by 5 and 61 modulo 612 and are all 1 mod 4, so neither trigger fires for any $d \mid 153$; full conductor excludes 305 because $F(612, 305) = 204 < 612$. This separates the displayed tests only; 305 was excluded long before.

Proposition 2.2 of [8] assumes a splitting relative difference set, and the Turyn lift in $C_{4t}$ is nonsplitting, so it does not apply. The 1994 erratum [7] replaces a scalar by an arbitrary group-ring coefficient in the 1991 proof [6], repairs parts (a),(b), rejects equation (11) and doubts part (c). The size bound above was reproved with arbitrary coset coefficients. We have not matched every part of the 1991 theorem to the later statements.

### 8.4 Global inversion and forbidden multipliers

The inverse-multiplier theorems of [5] do not imply T4 or the $C_3$-inversion theorem. In the lift to $C_4 \times G$, row symmetry acts by $(z, g) \mapsto (z, -g)$, whereas a global inverse multiplier acts by $(z, g) \mapsto (z^{-1}, -g)$. Partial inversion of a $C_3$ factor and the square-fibre actions of T4 also fix $z$. Theorems 4.2 and 6.5 of [5] in fact forbid a global inverse multiplier for a proper Turyn lift, so applying them to row symmetry would wrongly exclude prime powers. Its Theorem 5.3 uses

character values, a Galois sign change, coset constancy and coefficient rigidity, which is close to the last step of our $C_3$ proof; it does not supply the symmetry-transfer steps that precede it. Miller's forbidden-multiplier theorem [17] concerns ordinary difference sets and an order-$p$ multiplier on a $p^e$ component with $e \geq 2$, a different object and action.

### 8.5 Sources we could not check

The divisible-difference-set theorem of Arasu–Ma [11] is the most important unchecked comparison: the publisher returned HTTP 403 and the author repository holds only an abstract. The full 1991 Mann paper, some field-descent sources and several textbooks were not fully examined, and the mod-16 reconstruction has had only a preliminary prior-art search. These gaps limit every "new" label in Table 2. Priority claims of [3] are inherited, not re-examined here.

## 9 Coverage, certificates, and reproducibility

The finite claims use the range $5 \leq q < 10000$, $q \equiv 1 \pmod 4$. The eligibility restriction removes prime powers and parameters failing the two-squares condition; it is not a density among all integers. Table 3 records the disjoint dispositions and the specified selected union.

| Disposition or comparison | $q < 2000$ | $q < 10000$ |
|---|---|---|
| All parameters $q = 5, 9, \ldots$ | 499 | 2499 |
| Prime powers: existence known | 164 | 642 |
| Excluded by the two-squares condition | 211 | 1194 |
| Eligible non-prime-power two-squares parameters | 124 | 663 |
| Excluded by the selected reusable-criterion union | 122 | 644 |
| Additional recorded exclusions outside that union | 2 | 11 |
| Total eligible parameters excluded | 124 | 655 |
| Unresolved unconditionally in this record | 0 | 8 |

Table 3: Reconciled coverage. The selected-union and additional-exclusion rows partition the eligible exclusions; neither is added again to the total.

The selected union has eleven members, named as in `data/coverage.csv`: O (Theorem 2.2), FC (Theorem 3.2), T1 (Theorem 2.5, not its quotient corollary), T2 (Theorem 2.3), T4 (Theorem 2.7 with D1), T5 (Theorem 2.8), QE (Theorem 4.1), C2_uniform (Theorem 5.3), Full_prime_fibre (Corollary 5.5), Fixed_fibre (Corollary 4.4) and Doubling_origin (Theorem 4.3). These overlap. In particular, their individual counts cannot be added. The older bounded three-fibre enumeration is superseded by the uniform theorem, rather than being presented as its proof.

*Proof of Theorem 1.1.* The exact ledger contains one row for every $q = 5, 9, \ldots, 9997$. Its factorizations determine the prime-power and two-squares predicates. The verification script in `verify/` recomputes all eleven criteria from the statements above at every parameter. It finds exactly 644 eligible exclusions, agrees with every flag in `data/coverage.csv`, and excludes no prime power. The 19 eligible survivors of this selected union are

$$909, 1845, 2561, 2665, 3965, 4745, 5185, 5525, 6137, 6409,$$
$$7461, 7565, 7685, 7929, 8245, 8865, 8905, 9245, 9265.$$

Table 4 supplies recorded exclusions for eleven of these. Removing those eleven leaves exactly the eight stated parameters. The source census, baseline evidence and this join are separately pinned in the accompanying materials. The accompanying check script refactors every parameter and verifies the set difference and all displayed counts; it does not re-prove the earlier certificates. □

| $q$ | Additional recorded argument | Material |
|---|---|---|
| 909 | Full-unit $C_7$ Gaussian augmentation-ray test | M05–M06 |
| 1845 | Classical self-conjugacy obstruction | M06 |
| 2561 | Complete $C_{61}$ parity/augmentation census; separately colored factor matching | M07 |
| 2665 | Complete quotient-correlation spectrum | M06 |
| 6137 | Full-unit $C_{11}$ Gaussian augmentation-ray test | M05–M06 |
| 7461 | Full-unit $C_7$ Gaussian augmentation-ray test | M05–M06 |
| 7929 | Recorded degree-two period predicate at $C_{13}$ | M06 |
| 8865 | Forced factors, small-norm rigidity and disjoint augmentation residues at $C_{31}$ | M02 |
| 8905 | Incompatible complete $C_{61}$ and $C_{73}$ augmentation spectra | M02–M03, M07 |
| 9245 | Full-unit $C_{23}$ Gaussian augmentation-ray test | M05–M06 |
| 9265 | Complete residual norm and augmentation check at $C_{113}$ | M02 |

Table 4: The eleven exclusions outside the selected union. Several use general criteria; this is not a list of eleven isolated new results.

The baseline ledger distinguishes evaluated source predicates, exact enumerations and source theorem references. Its review reconstructed the lower-frontier cases and checked the broader bookkeeping, but did not rerun every expensive inherited solver above that frontier. That scope remains visible here, especially for the recorded 7929 predicate. The present coverage theorem relies on these credited source determinations as well as the later criteria. Not every row was recomputed for this paper.

### 9.1 Current unresolved parameters

| $q$ | $t = (q+1)/2$ | Factorization of $q$ | Status |
|---|---|---|---|
| 3965 | 1983 | $5 \cdot 13 \cdot 61$ | Open; stronger mod-16 reduction |
| 4745 | 2373 | $5 \cdot 13 \cdot 73$ | Open |
| 5185 | 2593 | $5 \cdot 17 \cdot 61$ | Open unconditionally |
| 5525 | 2763 | $5^2 \cdot 13 \cdot 17$ | Open |
| 6409 | 3205 | $13 \cdot 17 \cdot 29$ | Open |
| 7565 | 3783 | $5 \cdot 17 \cdot 89$ | Open |
| 7685 | 3843 | $5 \cdot 29 \cdot 53$ | Open |
| 8245 | 4123 | $5 \cdot 17 \cdot 97$ | Open |

Table 5: Parameters below 10000 left open by the arguments collected here. The earlier GRH-dependent result at 5185 remains separately recorded.

An order disappears from this table only when a complete argument applies to every hypothetical pair at that order. A timeout, a violated equation at one chosen row, or a failure of one relaxation witness is not sufficient. This is why the mod-16 work at 3965 does not change the table.

### 9.2 What the exact checks establish

For lattice certificates, exact prime-kernel membership and determinants identify the full ideal; exact inclusive-bound vector enumeration proves completeness. For quotient certificates, every allowed row or signed sum belongs to an explicitly finite domain. For the final coverage join, independently checked arithmetic flags and earlier certificates are combined by exact set operations. Corrupted inputs and known positive pairs were used in the supporting audits to test more than successful execution.

The file `data/coverage.csv` has one row per parameter, with its factorization, eligibility, status, firing criteria, material identifiers and the conditional note at 5185. The script `verify/verify_coverage.py` uses only the Python standard library and runs in about half a minute. Certificates for the eleven exclusions of Table 4 are in `certificates/`. The file `audit/README.md` lists the material identifiers M00–M17 with SHA-256 hashes of the research files they denote; those files are not distributed.

## A Retained auxiliary results and limits

The following results are proved in the materials (M12) but not used in the main text. They are not additional rows of coverage unless the selected census explicitly uses them.

**General subgroup averaging (T3).** For $H \le G \le (\mathbb{Z}/m\mathbb{Z})^*$ with $[G : H]$ odd, let $G_e$ be its image modulo $e \mid m$, and put $\eta_{e,C} = \sum_{v \in C} \zeta_e^v$. An $H$-invariant projected row system with odd Gaussian coordinates at unit positions satisfies

$$\frac{m\sqrt{2}}{[G : H]} \le \sqrt{q} \sum_{e \mid m} \sum_{C \in (\mathbb{Z}/e\mathbb{Z})^*/G_e} |\eta_{e,C}|.$$

Average over $G$, use the odd orbit-sum lower bound at position one, and group Fourier inversion by these cosets. This contains the full-unit averaging argument and variants using only the odd-order part of a quotient unit group. The exact scope and correction to early firing claims are in the retained proof/audit pair.

**Squarefree Jacobi-kernel extension (I2m).** For squarefree $m \mid t$, $(m, h) = 1$, $h = t/m$, and $w = \omega(m)$, invariance under the kernel of the Jacobi character on the full $m$-fibre necessarily gives

$$(h + 1)(m - 2^{w+1} + 1) \le 4h(2^w - 1)^2 + 2.$$

The character component supplies a support identity analogous to T4, and the identity component supplies the energy bound. Violation is sufficient for exclusion; the statement is not an equivalence between this inequality and pair existence. The full proof, including its large-$w$ induction, is retained in M12.

**A large-prime index-four theorem (Q4).** In the coprime split $t = hp$, with $h \ge 3$ odd, $p$ prime, $4 \mid p - 1$ and $f = (p - 1)/4$, coefficientwise invariance under the index-four subgroup is impossible if

$$p \ge 2h + 1, \qquad f(f - 1) > 2h, \qquad 2h < (f - 1)(h - 1).$$

For $h \ge 11$, the first inequality implies the other two. The source proof uses the exact quotient identity, sparse exceptional supports and two applications of Frobenius parity. Its corrected primitive

congruence must be combined with the quotient congruence modulo eight; the primitive equation alone is insufficient. This theorem supplies no extra current frontier closure in the recorded census.

**Limits and refuted shortcuts.** The source record retains the following negative findings because they prevent incorrect generalizations:

- The isolated quartic emptiness proposal has an explicit compatible Gaussian witness at 93721; it cannot support a nonexistence claim.
- Universal descent of all $C_{61}$ norm values to a quadratic subfield is false: 480 of the 504 norm-2561 values lie outside that quadratic subfield.
- Extending the three-primary cofactor threshold naively to every prime is false; the known pair at $t = 25, q = 49$ supplies a counterexample.
- A finite-prefix match of factor catalogs, a feasible quotient or a mod-16 solution is a necessary-condition witness, not an exact pair.
- At 3965 the specified degree-two certificate family has 803 rows in 19901 squarefree monomial columns, rank 795, and no constant one in its span. This does not establish nonlinear feasibility. The quadratic parts of its 199 original equations have rank 199.
- The proposed relative-$\mathbb{F}_4$ trace obstruction is refuted at $p = 101$. The stronger odd-support conjecture remains open.

The precise next generalization target is the compatibility of the forced inner fibre with both residual equations, not a larger search over an already feasible weaker relaxation. The explicit current formulations, counterexamples and unresolved weighted-carry dependency are indexed in M17.

# B Evidence map

The identifiers below denote the research files behind each result. They are not distributed, except for the certificates in `certificates/`; `audit/README.md` gives their SHA-256 hashes.

| ID | Subject | Evidence and scope |
|---|---|---|
| M00–M01 | Published baseline; foundations | Original source and initial mathematical dependencies; later records supersede earlier status labels. |
| M02–M03 | Small norms; 8865, 9265, 8905 | Proofs, residual classifications and the complete compatible-augmentation contradiction. |
| M04–M06 | Local arithmetic; prime powers; ledger | Proofs, exact quotient checks, baseline bookkeeping and explicit inherited-source limits. |
| M07 | Reusable $C_{61}$ and factor matching | General lattice cutoff; complete 2561 certificate; separate colored-factor verification. |

| ID | Subject | Evidence and scope |
|---|---|---|
| M08 | Arithmetic catalog transport | Universal proof, class-field sources and one independently checked ray-member certificate. |
| M09–M10 | Averaging and full conductor | Proofs, exact comparisons, kernel theorem and cofactor threshold. |
| M11 | Mann and subsequent sources | Erratum interpretation and direct prior-theorem substitutions, with access limits. |
| M12–M13 | Prime-fibre generalizations | Corrected source proofs, audits and later independent arithmetic reconciliation. |
| M14–M15 | Quotient energy and antecedents | Six finite closures, corrected counting formulas and superseded bounded three-fibre work. |
| M16 | Abelian three-fibre theorem; final census | Full proof and audit; 6437/7625 certificates; exact final criterion union and eight-case frontier. |
| M17 | Mod-16 reconstruction and latest source | Full proof/audit; 3965 fixed-outer certificate; independent $p = 101$ census; AJP1990 comparison. |

## References


[1] R. J. Turyn, Character sums and difference sets, *Pacific J. Math.* **15** (1965), 319–346.

[2] R. J. Turyn, An infinite class of Williamson matrices, *J. Combin. Theory Ser. A* **12** (1972), 319–321.

[3] J. Kline, Toward a converse of Turyn's theorem: obstructions to Turyn pairs at composite orders, version 0.2.0 (2026), doi:10.5281/zenodo.21776269.

[4] P. Delsarte, J. M. Goethals and J. J. Seidel, Orthogonal matrices with zero diagonal. II, *Canad. J. Math.* **23** (1971), 816–832. https://doi.org/10.4153/CJM-1971-091-x.

[5] K. T. Arasu, D. Jungnickel and A. Pott, Divisible difference sets with multiplier $-1$, *J. Algebra* **133** (1990), 35–62. https://www.sciencedirect.com/science/article/pii/002186939090067X.

[6] K. T. Arasu, D. Jungnickel and A. Pott, The Mann test for divisible difference sets, *Graphs Combin.* **7** (1991), 209–217. https://doi.org/10.1007/BF01787628. Full theorem and proof not obtained in the bounded comparison.

[7] K. T. Arasu, D. Jungnickel and A. Pott, Erratum: The Mann test for divisible difference sets, *Graphs Combin.* **10** (1994), 207. https://doi.org/10.1007/BF02986666.

[8] K. T. Arasu, D. Jungnickel, S. L. Ma and A. Pott, Relative difference sets with $n = 2$, *Discrete Math.* **147** (1995), 1–17. https://doi.org/10.1016/0012-365X(94)00226-9.

[9] D. Jungnickel and A. Pott, Perfect and almost perfect sequences, *Discrete Appl. Math.* **95** (1999), 331–359. https://doi.org/10.1016/S0166-218X(99)00085-2.

[10] K. T. Arasu and N. J. Voss, Answering a question of Pott on almost perfect sequences, *Des. Codes Cryptogr.* **18** (1999), 7–10. https://doi.org/10.1023/A:1008316615026. The prior-result identification uses the inspected abstract and bibliographic record; the complete proof was not reaudited.

[11] K. T. Arasu and S. L. Ma, A nonexistence result on difference sets, partial difference sets and divisible difference sets, *J. Statist. Plann. Inference* **95** (2001), 67–73. https://www.sciencedirect.com/science/article/pii/S0378375800002780. Metadata and abstract inspected; full DDS theorem unavailable in this comparison.

[12] B. Schmidt, Cyclotomic integers and finite geometry, *J. Amer. Math. Soc.* **12** (1999), 929–952.

[13] K. H. Leung and B. Schmidt, The field descent method, *Des. Codes Cryptogr.* **36** (2005), 171–188. Inspected author version: https://personal.ntu.edu.sg/bernhard/Publications/pub/field.pdf.

[14] M. M. Tan, Group invariant weighing matrices, arXiv:1610.01914v4 (2018). https://arxiv.org/abs/1610.01914v4. The cited definition and theorem numbers refer to this inspected revision.

[15] K. T. Arasu, D. M. Gordon and Y. Zhang, New nonexistence results on circulant weighing matrices, *Cryptogr. Commun.* **13** (2021), 775–789. https://doi.org/10.1007/s12095-021-00492-0. Inspected preprint: https://arxiv.org/abs/1908.08447v3.

[16] D. M. Gordon, Cyclic relative difference sets and circulant weighing matrices, *J. Algebraic Combin.* **63** (2026), Article 24. https://doi.org/10.1007/s10801-026-01515-w. Inspected preprint: https://arxiv.org/abs/2501.14924v2.

[17] N. Miller, Forbidden multipliers in abelian difference sets, arXiv:2511.10231v1 (2025). https://arxiv.org/abs/2511.10231v1.

[18] M. M. Tan, Character and multiplier obstructions for circulant weighing matrices, arXiv:2608.18468v2 (2026). https://arxiv.org/abs/2608.18468v2. Used as a recent mechanism comparison, not as an independently audited dependency of the present proofs.

[19] A. V. Sutherland, *18.785 Number Theory I*, Lecture 28, Massachusetts Institute of Technology, Fall 2021, Section 28.4, Theorems 28.6 and 28.9, and Remark 28.12. https://math.mit.edu/classes/18.785/2021fa/LectureNotes28.pdf.